

Honeywell

TPS Builder
Installation, Implementation, and
Administration
Guide

BL08-421

R421

03/2013

Notices and Trademarks

Copyright 2013 by Honeywell International Srl.

Release 421 03/2013

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special, or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell is a registered trademark of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell Process Solutions

1860 W. Rose Garden Lane

Phoenix, AZ 85027 USA

1-800 822-7673

About This Document

This document provides information for the following Honeywell TPS Builder configuration tool activities:

- Installation
- Implementation
- Basic TPS Builder concepts
- Database administration

Guidelines as well as detailed procedures are included.

TPS Builder can be installed on a variety of platforms, such as the Honeywell Global User Station (GUS), Application Processing Platform (APP) node, a stand-alone personal computer, or a network. This guide is intended for individuals who are familiar with the platform on which TPS Builder will be installed and used. This includes familiarity with the computer, its operating system, and networks when applicable.

This guide is also intended for users with administrative rights and experience, in the case of certain installation and administration procedures.

Release Information

Document Name	Document ID	Release Number	Publication Date
Installation, Implementation, and Administration Guide - BL08	BL08-421	421	03/2013
Document Category			
Installation and Upgrades			

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
TPS Builder Configuration Tool Guide	BL20
R421 03/2013	TPS Builder Installation, Implementation, and Administration Guide Honeywell

iii

Support and Other Contacts

United States and Canada

Contact: Honeywell Solution Support Center
Phone: 1-800-822-7673
Calls are answered by dispatcher between 6:00 am and 4:00 pm Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.
Fascimile: 1-973-455-5000
Mail: Honeywell TAC, MS L17
1860 W. Rose Garden Lane
Phoenix, AZ, 85027 USA

Europe, Middle East, and Africa (EMEA)

Contact: Honeywell TAC-EMEA
Phone: +32-2-728-2345
Fascimile: +32-2-728-2696
Mail: TAC-BE02
Hermes Plaza
Hermeslaan, 1H
B-1831 Diegem, Belgium
Email: tac-be02@honeywell.com

Pacific

Contact: Honeywell Global TAC – Pacific
Phone: 1300-364-822 (toll free within Australia)
+61-8-9362-9559 (outside Australia)
Fascimile: +61-8-9362-9564
Mail: Honeywell Limited Australia
5 Kitchener Way
Burswood 6100, Western Australia
Email: GTAC@honeywell.com

India

Contact: Honeywell Global TAC – India
Phone: +91-20- 6603-9400
Fascimile: +91-20- 6603-9800
Mail: Honeywell Automation India Ltd
56 and 57, Hadapsar Industrial Estate
Hadapsar, Pune –411 013, India
Email: Global-TAC-India@honeywell.com

Korea

Contact: Honeywell Global TAC – Korea
Phone: +82-80-782-2255 (toll free within Korea)
Fascimile: +82-2-792-9015
Mail: Honeywell Co., Ltd
4F, Sangam IT Tower B4-4 Block
1590, DMC Sangam-dong, Mapo-gu,
Seoul, 121-835, Korea
Email: Global-TAC-Korea@honeywell.com

People's Republic of China

Contact: Honeywell Global TAC – China
Phone: +86- 21-2219-6888
800-820-0237
400-820-0386
Mail: Honeywell (China) Co., Ltd
33/F, Tower A, City Center, 100 Zunyi Rd.
Shanghai 200051, People's Republic of China
Email: Global-TAC-China@honeywell.com

Singapore

Contact: Honeywell Global TAC – South East Asia
Phone: +65-6823-2215
Fascimile: +65-6445-3033
Mail: Honeywell Private Limited
Honeywell Building
17, Changi Business Park Central 1
Singapore 486073
Email: GTAC-SEA@honeywell.com

Japan

Contact: Honeywell Global TAC – Japan
Phone: +81-3-6730-7160
Fascimile: +81-3-6730-7228
Mail: Honeywell Japan Inc.
New Pier Takeshiba, South Tower Building,
20th Floor, 1-16-1 Kaigan, Minato-ku,
Tokyo 105-0022, Japan
Email: Global-TAC-JapanJA25@honeywell.com

Elsewhere

Call your nearest Honeywell office.

World Wide Web

Honeywell Solution Support Online:

<http://www.honeywellprocess.com>

Training Classes

Honeywell Automation College:

<http://www.automationcollege.com>

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Contents

1.	Introduction	15
1.1	Overview	15
	Welcome	15
	Document scope	15
	Intended audience.....	15
1.2	Getting Started	16
	TPS Builder fundamentals.....	16
	What do you want to do?.....	16
1.3	Technical Assistance	17
	If you need assistance	17
	Support Online	17
	Telephone support	17
	International customers	17
	Customers inside the United States	17
	Arizona customers	17
	Services provided.....	18
	Reporting a problem.....	18
2.	Installation Planning.....	19
2.1	Overview	19
	What is provided for TPS Builder installation?	19
	TPS Builder components.....	20
	Before you install TPS Builder.....	21
2.2	Basic Installation Requirements	22
	TPS networks.....	22
	Installing on a GUS Node for use with Fieldbus / Non-Fieldbus	22
	Installation configurations.....	23
	Who should install TPS Builder	25
	System requirements	25
2.3	Hard Drive/Server Space Requirements	27
	Considerations	27
	Summary.....	27
	Basic application requirements	27
	Database requirements.....	28
	Temporary files	28
2.4	Third-Party Software Requirements	29
	Considerations	29
	Software provided with TPS Builder.....	29
	User-supplied software	30
2.5	Planning Checklist	31

Contents

3.	Software Installation	33
3.1	Order of Activities	33
3.2	Migration.....	34
	Goal.....	34
	Reference.....	34
3.3	Installing TPS Builder on a GUS Node for Fieldbus / Non Fieldbus	35
	Use of TPN Server for Fieldbus/Non Fieldbus.....	35
	Installing TPS Builder on a GUS Node that is part of a TPS Domain	35
	Installing TPS Builder on a GUS Node that is not part of a TPS Domain	38
	TPS Applications that must be installed	39
	Upgrading TPS Builder on a GUS Node that is to remain part of a TPS Domain.....	44
	Installing HCI Client Connectivity application	45
	Upgrading TPS Builder on a GUS Node that will be removed from the TPS domain	47
3.4	SQL Server 2005 Installation	48
	What if SQL Server is already installed?	48
	Where to install SQL Server.....	48
	SQL Server Software	49
	SQL Server installation preparation.....	49
	Installation procedure	49
	Establish network user group	55
3.5	CodeWright Installation	56
	Before Installation.....	56
	Installation Procedure.....	56
3.6	TPS Builder Installation	57
	Where to install TPS Builder.....	57
	CD-ROM containing TPS Builder	57
	TPS Builder installation preparation	57
	Installation problems	63
3.7	TPS Builder Folder Structure	65
	TPS Builder repositories.....	65
	"Program Files" contents	66
3.8	Device Save/Restore Installation	67
	Functional description	67
	CD-ROM containing Device Save/Restore.....	67
	Where to install Device Save/Restore	67
	Requirements for the Save/Restore function.....	67
	Device Save/Restore installation procedure.....	68
4.	Post-Installation Activities.....	71
4.1	Post-Installation Additions	71
	Types of additions	71
	User-supplied software.....	72
4.2	Post-Installation Addition of TPS Builder Packages	73

Description	73
Procedure.....	73
4.3 Post-Installation Addition of CL Editor	74
Procedure.....	74
4.4 Uninstall TPS Builder	75
Description	75
Remove TPS Builder application.....	75
Remove TPS Builder, including repository	76
4.5 Stop, Start SQL Server Service	76
4.6 Correct SQL Server Password	77
Problem.....	77
What happens	77
Solution	77
5. Software Configuration	79
5.1 Overview	79
Configuration considerations.....	79
Before configuring TPS Builder	79
What can be configured	79
5.2 Define How Projects and Libraries are Opened	80
Description	80
Procedure.....	80
5.3 Set Up the Workspace	81
Description	81
Libraries	81
Workspace layout.....	81
Save workspace layout	81
5.4 Set Up Composition Views	82
Fundamentals	82
Planning	82
Choose default view for new projects.....	82
Create a new view.....	83
Change a view name	84
Delete a view.....	85
5.5 Define Parameter Configuration Form Format	86
Options.....	86
Tabbed layout	86
Alphabetized layout.....	87
DEB-order layout.....	87
Read-only parameters.....	87
Print NIM priorities to .EB file	88
Choosing form options	89
5.6 Define Default Size for Drawing Symbols.....	90
How the default size works.....	90
Preparation.....	90

Contents

Procedure.....	90
5.7 Set Log Options.....	92
Description	92
Enable logging.....	92
Enable detailed audit.....	92
Manage log size	93
5.8 Set Fieldbus Options	94
Description	94
Use of placeholders.....	94
Error reporting	95
Device matching.....	95
5.9 Set Online Operations Data	96
Description	96
Preparation.....	97
Procedure.....	97
6. Database Administration	99
6.1 Overview.....	99
Using this information.....	99
Function Summary	99
Space availability.....	100
6.2 TPS Builder Files.....	100
Honeywell-supplied database files	100
User-created database files	100
Project files.....	101
Repositories	101
6.3 Backup TPS Builder Repository	102
Overview	102
When you should back up	102
Permissions.....	103
Procedure.....	103
6.4 Restore TPS Builder Repository	104
Overview	104
Permissions.....	104
Procedure.....	104
6.5 TPS Builder Distribution Options.....	106
Planning for distribution.....	106
Distribution options.....	106
6.6 Distributing Templates.....	107
Preparation.....	107
Distributing the database.....	107
6.7 Export TPS Builder Repository	108
Overview	108

Permissions.....	108
Procedure.....	108
6.8 Export TPS Builder Objects	109
Why exporting takes place	109
Transfer repository.....	109
What is exported?	110
What is not exported?	112
Plan the export	112
Export objects procedure	113
Cancel an export	115
Export problems	115
Working with folders in Export (Step 1)	116
6.9 Import TPS Builder Repository	117
Overview	117
Permissions.....	117
Procedure.....	117
6.10 Import TPS Builder Objects	119
Description	119
Transfer repository source	119
What is imported?	119
Terminology used.....	120
Results of template import.....	120
Results of strategy import.....	123
Results of library import.....	124
Results of project import.....	125
Import example	126
Plan the import	127
Import objects procedure.....	128
Cancel an import	131
Import problems	131
6.11 Evaluating Export and Import Details	132
Description	132
Evaluate source objects	132
Comparing transfer and target repositories	133
Comparison criteria	134
6.12 Migrate TPS Builder Repository	137
Overview	137
Permissions.....	137
Procedure.....	137
6.13 Create TPS Builder Repository.....	141
Overview	141
Permissions.....	141
Procedure.....	141
6.14 Delete TPS Builder Repository	142
Overview	142
Permissions.....	142
Procedure.....	142
6.15 Switch TPS Builder Repository	143

Contents

Overview	143
Procedure.....	143
 7. TPS Builder API	 145
7.1 Overview.....	145
APIs.....	145
Intended use.....	145
Class hierarchy.....	145
7.2 Examples.....	146
Example use.....	146
Example API script.....	146
7.3 Using the Script.....	148
Using the script with your data	148
Viewing the script	149
Creating strategies from external data	150
 8. Maintaining Adequate Disk Space	 151
8.1 Overview.....	151

1. Introduction

1.1 Overview

Welcome

The Honeywell TPS Builder is an easy-to-use tool for graphical control strategy building. Its user-friendly interface, use of exposure rules, and many other functions provide efficiency. Its strong parameter checking and data flow connection checking help reduce configuration errors.

TPS Builder combines the strong traits of a “single point builder” with the power of a “bulk builder.” Bulk building involves the use of standard Microsoft Excel functions to quickly build similar points, and to edit parameters for multiple points at the same time.

Templates also play a key efficiency role—providing the foundation for all configuration data entry. You can create your own templates to streamline configuration, and create templates of complete strategies (not just individual points).

Document scope

This document provides information for the following Honeywell TPS Builder configuration tool activities:

- Installation
- Implementation
- Basic TPS Builder concepts
- Database administration

Guidelines as well as detailed procedures are included.

Intended audience

TPS Builder can be installed on a variety of platforms, such as the Honeywell Global User Station (GUS), Application Processing Platform (APP) node, a stand-alone personal computer, or a network. This guide is intended for individuals who are familiar with the platform on which TPS Builder will be installed and used. This includes familiarity with the computer, its operating system, and networks when applicable.

This guide is also intended for users with administrative rights and experience, in the case of certain installation and administration procedures.

1.2 Getting Started

TPS Builder fundamentals

After installing TPS Builder (as explained in this document), but before using this tool, read the *TPS Builder Configuration Tool User's Guide*, *TPS Builder Fundamentals* to gain an understanding of basic TPS Builder concepts.

What do you want to do?

Choose an area of interest:

If you want to...	... And your operating system is...	Then go to...
Plan for TPS Builder installation	Windows XP	This manual: : Installation Planning, Overview
Install TPS Builder for the first time	Windows XP	This manual: Software Installation, Installation Planning
Configure TPS Builder software	Windows XP	Software Configuration, Overview This manual: Installation Planning, Software Installation
Migrate an existing TPS Builder installation to this release	Windows XP	<i>TPS Builder Software Change Notice</i> (paper document provided with TPS Builder software)
Manage your TPS Builder database	Windows XP	Database Administration
Use TPS Builder	Windows XP	<i>TPS Builder User's Guide</i>
Obtain technical support	Windows XP	Technical Assistance Center

1.3 Technical Assistance

If you need assistance

If you need technical assistance, contact your local Honeywell Service Organization, as explained in the following paragraphs. For most efficient assistance, obtain the information requested in “Reporting a problem” before contacting Honeywell.

Support Online

Total Plant Support Online is Honeywell's global online support facility. Comprehensive support information and services are provided for Industrial Automation and Control (IAC) group products, systems, vertical market specific applications, and other IAC technologies.

Learn about Total Plant Support Online at:
<http://www.iac.honeywell.com/Pub/Mktg/support.stm>.

Telephone support

Obtaining telephone support depends on your location. The following paragraphs explain various options, depending on your location.

International customers

Outside of the United States, contact your local Honeywell Service Organization. If you are not sure of the location or telephone number, call your Honeywell representative for information.

Customers inside the United States

Within the United States, call the Honeywell Solution Support Center's Technical Assistance Center (TAC) at the toll-free number 800-822-7673.

Arizona customers

Within Arizona, the local number for TAC is 602-313-5558.

1. Introduction

1.3. Technical Assistance

Services provided

Calls to TAC are answered by a dispatcher from 7:00 A.M. to 5:00 P.M., Mountain Standard Time (6:00 A.M. to 4:00 P.M. when daylight saving time is in effect). Outside of these hours, emergency calls—those that affect your ability to control or view a process—will be received by an answering service, and returned within one hour.

TAC maintains its own **TotalPlant** Solutions system, and frequently can duplicate problems on this equipment.

Reporting a problem

Describe your problem with as much detail as possible. Provide the following information to the extent possible:

- System configuration
- Is the problem reproducible? Do you get the problem regardless of the type of computer you are using?
- In which context does the problem occur? That is,
 - What were you doing?
 - Is the sequence of events important?
 - Does it make a difference when one or many computers are connected?
- Provide information from applicable log files. (From the TPS Builder main menu, choose Reports > Logs, and print the applicable part of the log.) For details on TPS Builder logs, refer to the TPS Builder Configuration Tool User's Guide

2. Installation Planning

2.1 Overview

What is provided for TPS Builder installation?

The R141 and later releases of TPS Builder are packaged on a separate DVD that contains all software necessary to support TPS Builder. Components necessary for installing TPS Builder are listed in the following table.

Honeywell

- TPS Builder, including:
 - the basic application
 - all available subsystems/options
 - the database administration tool
 - TPN Server (Installed from TPS System Software CD)
 - CL Editor
- Device Save/Restore



TIP

TPN Server is required if you are using a GUS station

Microsoft

- Redistributables
- SQL Server

Other

- Borland Corporation's CodeWright, which is required for use of CL Editor.
- CD Browser application

2. Installation Planning

2.1. Overview

Documentation

- Knowledge Builder CD-ROM, which includes specific documentation for TPS Builder.
- This document provides information about – Instructions on installing TPS Builder and related components; information on setting up the TPS Builder work environment; details on working with the TPS Builder repository; information on the TPS Builder Application Programming Interface (API).
- *System Management Configuration Guide (EX-08)* for systems using System Management components.
- *TPS System Implementation Guide (TP08X)* for Windows XP operating systems.
- Migration instructions and important details about the product's release.
- TPS Builder license requirement and explanation. Explanation of your license to use TPS Builder.

TPS Builder components

TPS Builder itself includes the following components:

- The basic application
- Subsystems: Libraries for configuration of various parts of a TotalPlant Solution system, such as:
 - Local Control Network/Universal Control Network (including Fieldbus)
 - Data Hiway
- Options: Control Drawing-to-GUS Export Tool add-on allows you to export TPS Builder drawings to GUS Display Builder (.pct file).
- Database Administration Tool: Allows you to manage the SQL Server database.
- CL Editor: Control Language (CL) Editor allows you to efficiently write CL programs using Borland CodeWright.
- Device Save/Restore: Used to save and restore the configuration of Fieldbus devices within a TPS Network.

Required for, and provided with, TPS Builder is Microsoft SQL Server, and Borland CodeWright 7.5.

Required for and provided with TPS system software CD is TPN Server. It is used as an HCI server; allows you to do download Fieldbus / Non Fieldbus configurations from a GUS workstation.

The installation of these components is covered in this section.

Before you install TPS Builder

Several considerations will help ensure a successful TPS Builder installation. They include:

Consideration	Discussion
Basic Installation Requirements	Deciding who will install the software, and the system(s) on which TPS Builder will be installed.
Hard Drive/Server Space Requirements	Calculating the amount of system and/or server drive space required for TPS Builder and its operation.
Third-Party Software Requirements	Understanding the relationship of TPS Builder to third-party software, and ensuring this software is installed as required.
Planning Checklist	This checklist provides specific steps regarding the previous planning stages.

Each of these considerations is covered in the remainder of this section.

2. Installation Planning

2.2. Basic Installation Requirements

2.2 Basic Installation Requirements

TPS networks

It is essential that the TPS network and the Global User Stations (GUSs) are at the proper version and are properly configured. For details on ensuring the TPS network is configured as required, refer to:

- the Software Change Notice
- the *System Management Configuration Guide (EX-08)*
- the *TPS System Implementation Guide (TP08X)*

Installing on a GUS Node for use with Fieldbus / Non-Fieldbus

If you are installing TPS Builder for the first time on a GUS Node, the node may be configured as part of a TPS Domain, or it may be configured outside of a TPS Domain, in which case a Domain Controller (DC) is not required.

If you are installing TPS Builder on a GUS Node that is currently configured as part of a TPS Domain, you may choose to retain it in the domain or to remove it from the domain, in which case a DC is no longer required.

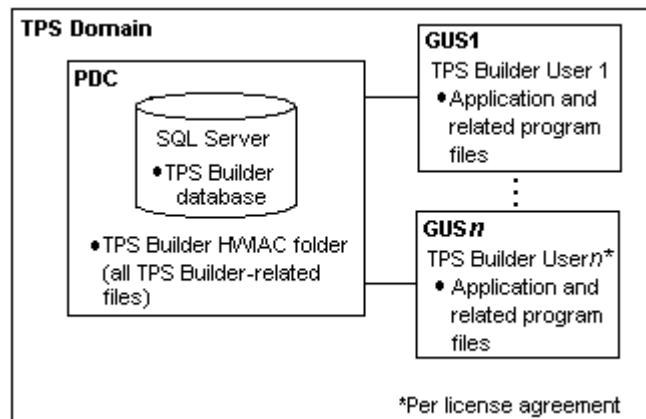
Additional requirements exist when TPS Builder is used for Fieldbus configuration.

Installation configurations

The three main installations for installing TPS Builder are introduced in the following paragraphs. Details are provided later in this section.

1. Plant Control Network: This installation includes installing TPS Builder components within a TotalPlant (TPS) Solutions domain that is part of a plant control network. This installation is required for download/upload of configuration data to TPS Network (TPN) or Fieldbus devices.

The following figure shows the basic TPS Builder installation within a TPS domain.



In this case,

- SQL Server must be installed on a domain controller (DC) within that domain.
- The TPS Builder application is installed on one or more Global User Stations (GUSs).

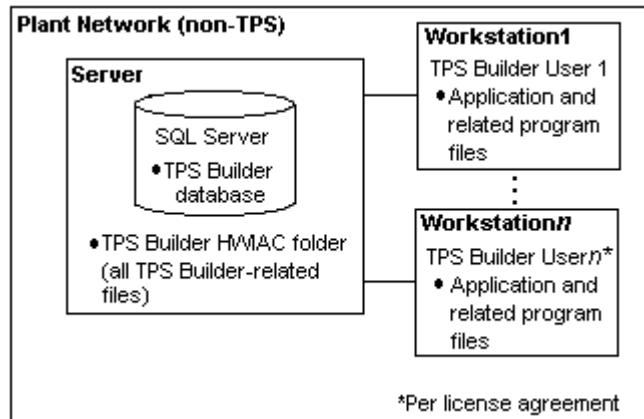
2. Other Networks: This installation includes installing TPS Builder components in a network that is separate from the plant control network and TPS domain. This installation would be for configuration purposes only, not download/upload. In this case:

- Install SQL Server on a server within the network.
- Install the TPS Builder application on one or more workstations within the network.

2. Installation Planning

2.2. Basic Installation Requirements

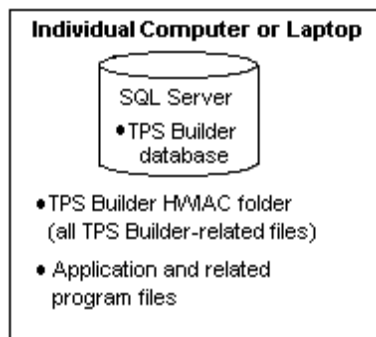
The following figure shows the network installation approach.



In this case, configuration data can be transferred to a GUS in the plant control network for subsequent downloading.

3. Stand-Alone Computer: Installing TPS Builder on a computer that is not part of a TPS system or a network, in which case everything can be installed on the same computer, as shown here.

In this case, configuration data can be transferred to a GUS in the plant control network for subsequent downloading.





ATTENTION

Before installing TPS Builder on a stand-alone computer, ensure that:

- the network connection is active.
 - the NIC card has a manual IP address.
-

Who should install TPS Builder

For network installations, the TPS system administrator or network administrator should install TPS Builder and its components.

If TPS Builder is to be installed on an individual computer or laptop, a user with administrative rights can perform the installation.

System requirements

Following are system requirements for installing and using TPS Builder.

As with most applications, the more advanced your system is, the better TPS Builder will run. (“More advanced” means faster processor, more RAM, bigger hard drive, more free space as opposed to less free space.)

Item	Requirement
All Installations	
DVD Drive	Any system on which TPS Builder or its components are to be installed must include a DVD Drive.
Display resolution	1024 x 768 pixels minimum
Hard drive space	Refer to “Hard Drive/Server Space Requirements” in this section.
Memory	1GB minimum
Microsoft components	The following must be installed: • Microsoft Redistributable Libraries It is a part of TPS Builder installation.

2. Installation Planning

2.2. Basic Installation Requirements

Item	Requirement
Operating system	- GUS: Microsoft Windows XP with Service Pack recommended by Honeywell. - Other PC: Windows XP with Service Pack recommended by Honeywell.
Processor	Intel microprocessor, Pentium Pro 200 MHz or higher recommended
Swap File Size	As recommended by Microsoft.
Control Network Installations	
Domains	TPS domain based on Windows 2003 server
Global User Station	- Intel version - R400 or later only
TPN Server	The TPN Server is installed on each GUS as part of the TPS installation and the installation is transparent to the user. The default access level is set to ViewOnly. Refer to the <i>TPN Server User's Guide</i> for details.
Other-Network Installations	
Network	For network installations, TPS Builder is certified on Windows XP and Service Pack as recommended by Honeywell.
For Fieldbus Configuration	
Fieldbus device download firmware and utility	Each release of TPS Builder requires specific versions of Fieldbus device download firmware and download utility. Refer to the TPS Builder <i>Software Change Notice</i> (paper document) for details related to the current TPS Builder release.
Fieldbus templates	Updates for Fieldbus device firmware may require updates to Fieldbus templates in TPS Builder, after TPS Builder is installed. Refer to the <i>Software Change Notice</i> for specific instructions.
TPN release	R681

Item	Requirement
NI Configurator	If the NI Configurator is to be used to download new versions of device software, it must <i>not</i> be installed on a GUS on which TPS Builder is being used.

2.3 Hard Drive/Server Space Requirements

Considerations

TPS Builder requires a substantial amount of hard drive/server space for the application components, its database, and for operational purposes. These requirements are discussed in the following paragraphs.

Summary

The following table presents a summary of all requirements. Details are presented in the following paragraphs.

Item	Free Space Required (MB)	
	Each application system	Database system
TPS Builder application	35	
SQL Server		675
TPS Builder database		40
Installation/basic use totals	35	230
TPS Builder database growth		50+*
15,000 strategies		81
40,000 strategies		118
*Depends on number of strategies		

Basic application requirements

For each system on which TPS Builder is installed, the minimum hard disk space required for installation is **35 MB**, for the following components.

2. Installation Planning

2.3. Hard Drive/Server Space Requirements

Element	Space Required (MB)
TPS Builder application (including libraries) and associated applications (CL Editor, Database Administration Tool, Device Save/Restore)	25
CL Editor, including CodeWright	10

Database requirements

Ensure that you have adequate space where the database will be installed, both for initial installation and growth of the database as you use TPS Builder. The initial installation requires nearly **200 MB**, as follows:

Element	Space Required (MB)
SQL Server	675
TPS Builder database (initial size)	40

Examples of TPS Builder database size after TPS Builder is in use:

Example	Space Used (MB)
Database containing 15,000 strategies	81
Database containing 40,000 strategies	118

Temporary files

As with any application, TPS Builder requires the use of temporary files. When calculating total free space, include sufficient free space to accommodate such files.

2.4 Third-Party Software Requirements

Considerations

TPS Builder and associated programs interact with additional software. Some of this software is provided with TPS Builder, and some is the responsibility of the user, as discussed in the following paragraphs. Specific installation of this software is discussed in this section.

Software provided with TPS Builder

The following table identifies third-party software that is provided with TPS Builder, and provides details about its role. These items must be installed as explained in the “Software Installation” section.

TPS Builder-Supplied Software	Comments
Microsoft • Redistributable Libraries • SQL Server 2005 Standard Edition with SP2	Redistributables are required on the system on which the TPS Builder application is installed. SQL Server is used for the TPS Builder database. SQL Server resides on the TPS Builder CD-ROM.
Borland CodeWright 7.5 or later	For using CL Editor. You cannot install CL Editor unless Borland CodeWright is installed.
Seagate Crystal Reports Runtime	For standard reporting. Used for the reports available within TPS Builder. It is embedded in TPS Builder and no installation is required.

2. Installation Planning

2.4. Third-Party Software Requirements

User-supplied software

The following table shows TPS Builder functions that require software that the user must provide.

Function	User-Supplied Software	Comments
Necessary software		
For bulk building and editing	Microsoft Excel 97,2000 or later	Microsoft Excel must be installed to use TPS Builder's bulk building and editing capabilities.
Optional software		
For custom reporting tasks	Seagate Crystal Reports Pro 5.0 or later	To perform advanced reporting tasks, obtain and install custom reporting software. Crystal Reports is preferred, because you can add reports to TPS Builder menus.
For Device Save/Restore	Microsoft Access 7.0 or later, or ODBC Data Source Administrator	To maintain the Device Save/Restore database, you can use this type of package to compact the database.
	Microsoft Access 7.0 or later, or any package capable of accessing an ODBC data source	For the Device Save/Restore database, you can use this type of package for the query/report capabilities, if desired.
To use the Application Program Interface (API)	Any OLE Automation controller, such as Microsoft Visual Basic 5.0 or Microsoft Excel 7.0 or later	To use the TPS Builder API in your own scripts and applications. Refer to the "TPS Builder API" section of this document.

2.5 Planning Checklist



ATTENTION

Refer to the *System Management Configuration Guide (EX-08)* and *TPS System Implementation Guide (TP08X)* for information relating to Windows 2003/XP and TPS Domain configuration.

Follow these steps before installing TPS Builder.

Step	Action	Comments
1	Determine installation locations.	Determine where all major components of TPS Builder are to be installed: • SQL Server • TPS Builder application References: • In this section, refer to – Basic Installation Requirements – Hard Drive/Server Space Requirements
2	Check system requirements.	Make sure the systems on which TPS Builder and components will be installed meet requirements. Reference: Basic Installation Requirements
3	Install user-supplied third-party software.	On each system on which TPS Builder application is to be installed, install user-supplied software (necessary and optional). (You may, however, install these programs after TPS Builder installation.) References: • Third-Party Software Requirements • Manufacturer's installation instructions
4	Install HCI Runtime	If you are installing Builder in a Domain controller of Windows XP/2003, you must install HCI Runtime from:- //remove this For Windows 2000 systems, HCI Runtime needs to be installed from Common Component CD. HCI Runtime needs to be installed from TPS System Software CD.

2. Installation Planning

2.5. Planning Checklist

Step	Action	Comments
5	If you will be using the CL Editor, install Borland CodeWright.	You cannot install/use CL Editor unless CodeWright is installed. Unless you already have CodeWright, you must install it. It is supplied with TPS Builder. References: • Software provided with TPS Builder • CodeWright installation instructions
6	Locate and read the following (paper) documents provided in the TPS Builder package: • License agreement • Software Change Notice (SCN)	The license agreement explains your product rights and provides confidential license and authorization numbers. You need these numbers during installation. The SCN provides details about the current TPS Builder release, including migration instructions.

3. Software Installation

3.1 Order of Activities

The order in which you install the various components required for TPS Builder is critical. You cannot install TPS Builder unless prerequisite items are installed. Follow the order presented in the following table.



ATTENTION

This information is for **NEW XP** installations or upgrades of an **R410** version only. If you are migrating from an earlier version of TPS Builder, refer to the *Software Change Notice*.

Step	Action	Comments
1	Complete planning and preparation steps.	Refer to the "Installation Planning" section. Note that while some of the third party software can be installed after TPS Builder and components are installed, it is preferable to install it first.
2	Uninstall earlier versions of TPS Builder.	To migrate to pre-R410 TPS Builder, refer to the <i>Software Change Notice</i> . To uninstall TPS Builder, refer to "Uninstall TPS Builder."
3	Install SQL Server 2005 Standard Edition.	Refer to "SQL Server 2005 Installation" in the Software Installation section.
4	Install GUS 400 or later (for TPS installation).	Refer to the appropriate GUS documentation. Upgrade instructions are contained in the <i>TPS Software Change Notice</i> .
5	Install CodeWright 7.5, if you intend to develop CL.	Refer to "CodeWright Installation" in the Software Installation section.

3. Software Installation

3.2. Migration

Step	Action	Comments
6	Install TPS Builder application. Important! For non-GUS installations, ensure the Microsoft Redistributables are installed here (part of TPS Builder installation).	Refer to “TPS Builder Installation” in the Software Installation section.
7	Install Device Save/Restore.	Refer to “Device Save/Restore Installation” in the Software Installation section.

3.2 Migration

Goal

The goal of migration is to bring the TPS Builder database, all of your projects, library customizations (specialized templates and/or user libraries), and drawings, forward into the current release.

Reference

Detailed migration instructions are provided in the (paper) *TPS Builder Software Change Notice*.

3.3 Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

This section discusses the different options available when installing TPS Builder on a GUS Node for use with Fieldbus/Non-Fieldbus.

- If you are installing TPS Builder for the first time on a GUS Node, this node may now be configured as part of a TPS Domain, or may be configured outside of a TPS Domain, in which case a DC is not required.
- If you are installing TPS Builder on a GUS Node that is currently configured as part of a TPS Domain you may choose to retain it in the domain or to remove it from the domain, therefore no longer requiring a DC.

The following paragraphs describe the process of Installing TPS Builder for the first time on a GUS Node, and also Upgrading from an existing version of TPS Builder.

Use of TPN Server for Fieldbus/Non Fieldbus

- The R400 and later release of TPS Builder needs TPN server to be installed from the TPS System Software CD.
- When you install TPS Builder R410 on a GUS Node the TPN Server will not be installed as part of the TPS Builder install.

Installing TPS Builder on a GUS Node that is part of a TPS Domain

This section describes requirements when this is the first time that TPS Builder has been installed on a GUS Node, which is part of a TPS domain.

3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

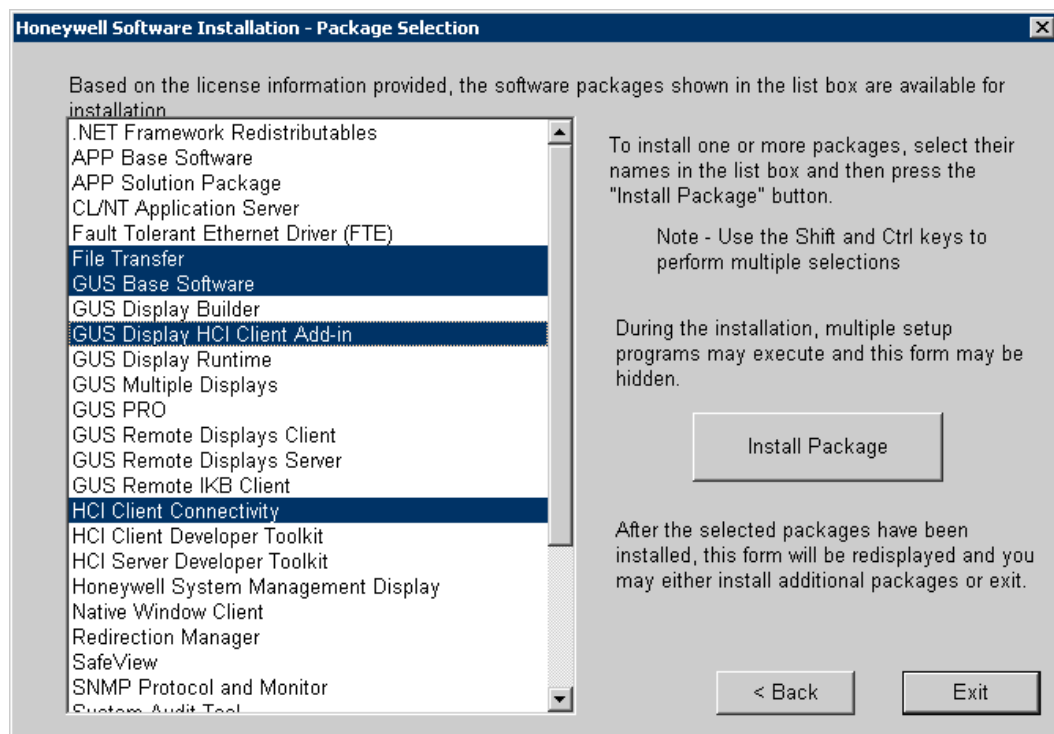


ATTENTION

Refer to the *System Management Configuration Guide (EX-08)* and *TPS System Implementation Guide (TP08X)* for information relating to Windows 2003/XP and TPS Domain configuration.

TPS Applications that must be installed (Windows XP only)

When you are installing TPS Builder on a GUS Node to allow you to perform downloads to Fieldbus Links, you must install a minimum set of TPS applications. These applications are shown in the following two dialog boxes taken from the TPS installation script.



3. Software Installation

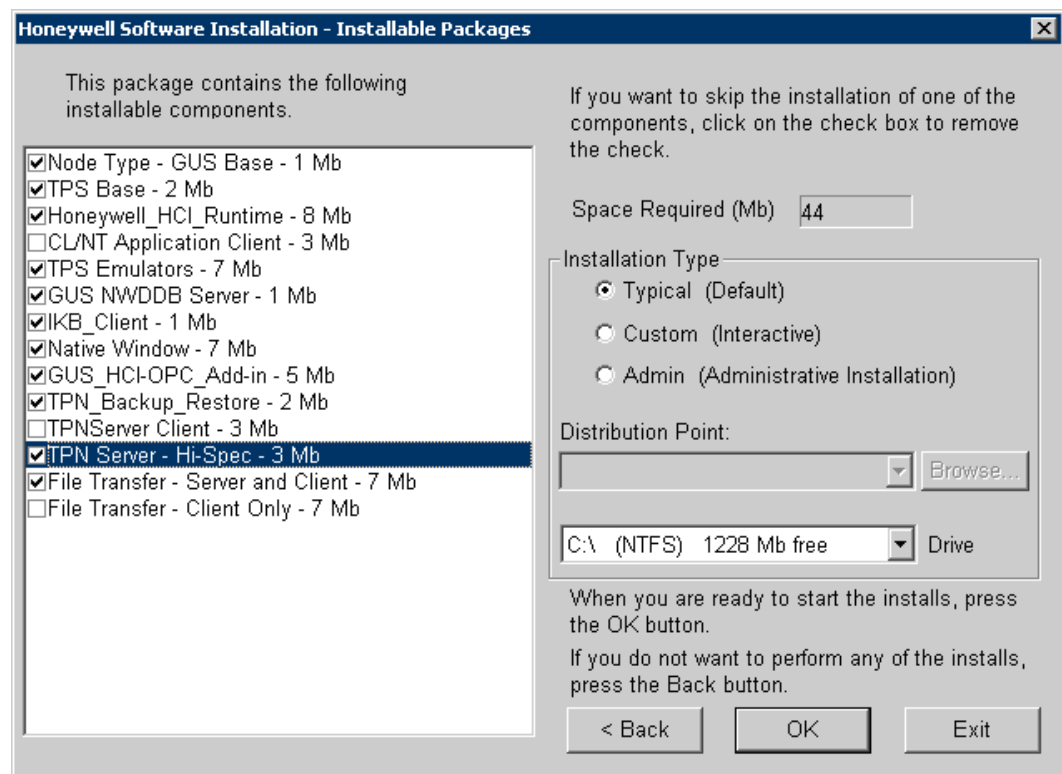
3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus



ATTENTION

TPN Server is one of the Packages selected. TPN Server is not installed as part of the TPS Builder install.

After you make the package selections and select the Install Package option, the following dialog box is displayed.



When the dialog box is first displayed all options are selected. Uncheck those shown above. Select OK. The selected TPS applications are installed.

3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

Installing TPS Builder on a GUS Node that is not part of a TPS Domain

This section describes requirements when this is the first time that TPS Builder has been installed on a GUS Node that is not part of a TPS domain.

If you are installing TPS Builder on a GUS Node that has never been part of a TPS Domain and you intend to run outside of a TPS Domain, you must perform the following before installing TPS Builder R410 or later.

- Install the minimum TPS applications. (see “TPS Applications that must be installed”)
- Uninstall any existing version of TPS Builder (see “Before you install TPS Builder”)

Special Considerations when the GUS Node is not part of a TPS Domain

If you are installing on a GUS Node that is not part of a Windows domain you must consider the following:

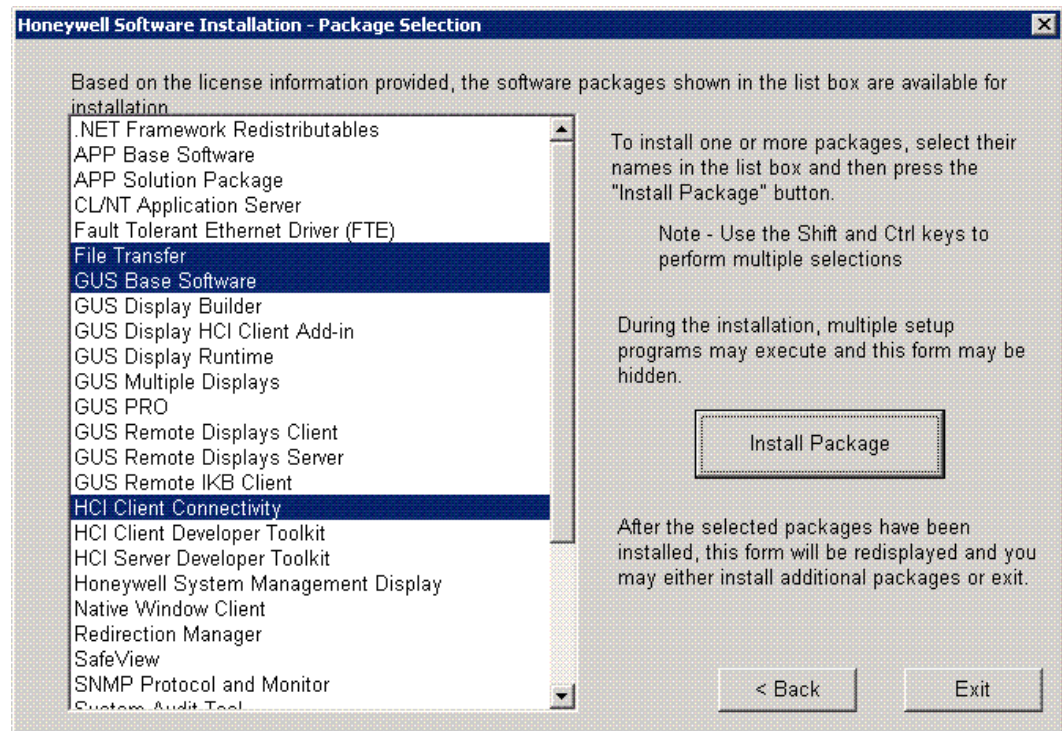
- ***Make the computer a member of a Workgroup.*** Use the Start/Control Panel/Network connections option, and select the General tab. Ensure that the LCNP card is connected to the LCN.
- ***Check the TCP/IP Protocol settings.*** Use the Start/Control Panel/Network connections > Local Area Connection, right mouse click and select the properties. In the General Tab select Internet Protocol (TCP/IP) > properties On the General tab, specify the 'Use the following IP Address' option and enter the IP Address details of the GUS workstation. To obtain the required information use IPCONFIG in a DOS window. On the General tab of Internet Protocol (TCP/IP) Properties dialog, click on the advanced option > WINS Address tab, ensure that the 'Enable DNS for Windows Resolution' and the 'Enable LMHOSTS Lookup' options are selected.
- ***Add the details of the workstation to the LMHOSTS file.*** An example of this file (named LMHOSTS.SAM is found at \Windows\system32\drivers\etc). Follow the instructions found in the document, *File Transfer Installation and Use - FT05200B - R200 - 5/99*, paragraph *PC-to-TPN Configuration* (page 16). Note also that you need to rename the file LMHOSTS and copy it to the \Windows\system32 directory. You then need to return to the TCP/IP Protocol dialog, WINS Address tab, and select the Import LMHOSTS... command and read the LMHOSTS file. If you already have an LMHOSTS file, you should add the details to the existing file.

3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

TPS Applications that must be installed

When you are installing TPS Builder on a GUS Node to allow you to perform downloads to Fieldbus Links, you must install a minimum set of TPS applications. These applications are shown in the following two dialog boxes taken from the TPS installation script.



3. Software Installation

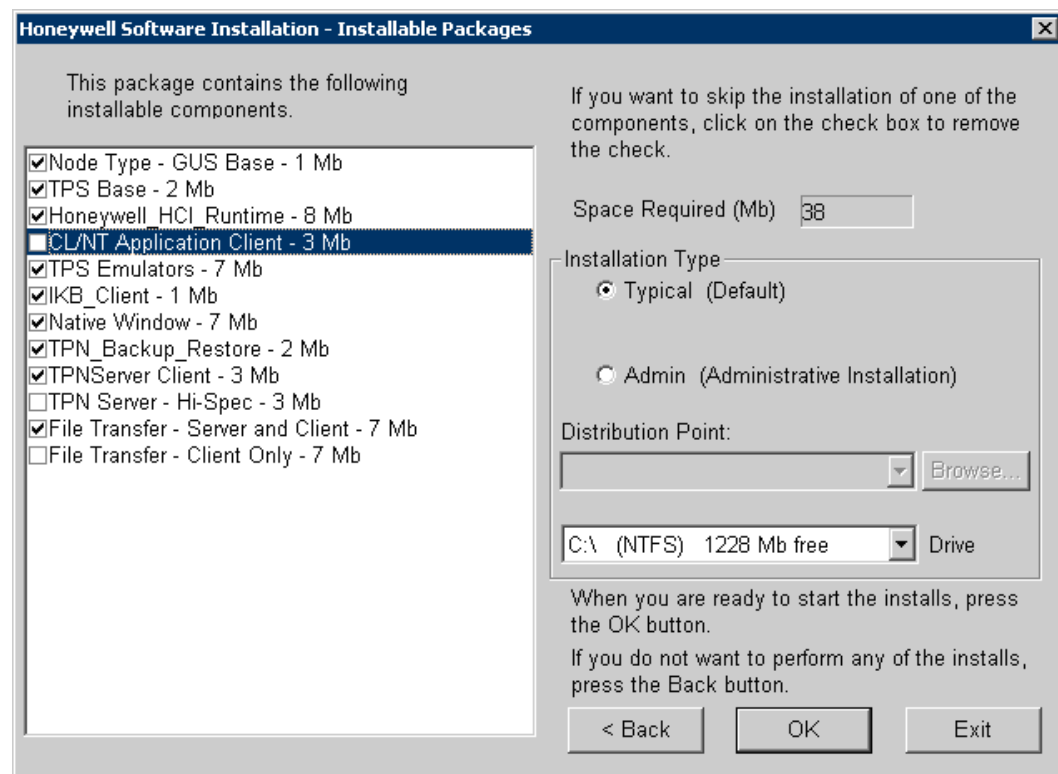
3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus



ATTENTION

TPN Server is one of the Packages selected. TPN Server is not installed as part of the TPS Builder install.

After you make the package selections and select the Install Package option, the following dialog box is displayed.



When the dialog box is first displayed all options are selected. Uncheck those shown above.

Create User TPSComServer

**ATTENTION**

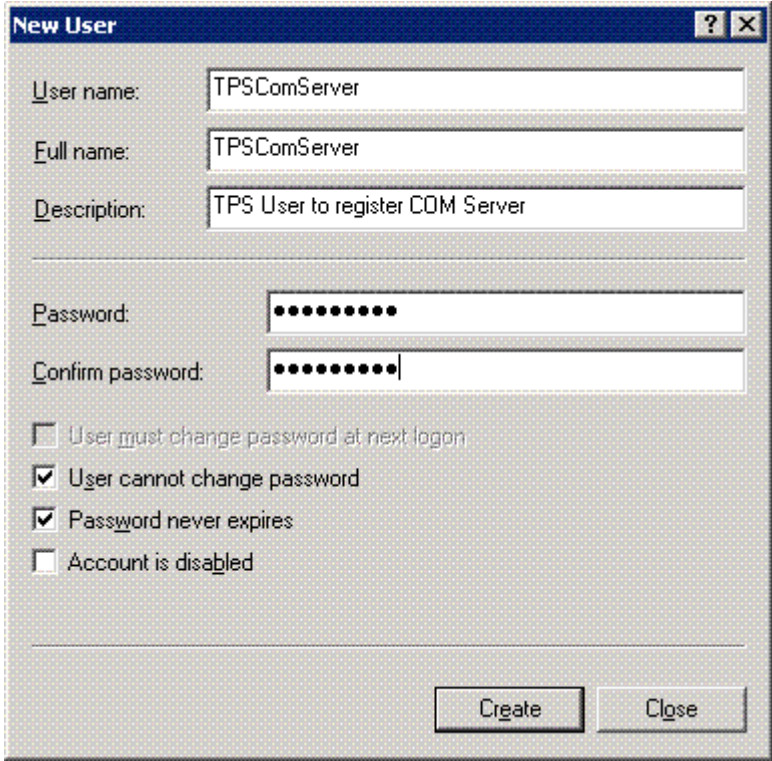
Refer to the *System Management Configuration Guide (EX-08)* and *TPS System Implementation Guide (TP08X)* for information relating to Windows 2003/XP and TPS Domain configuration.

As you are not installing certain TPS Applications, you must create a User that is normally created during the installation of these applications.

No	Action	Result
1	Select Start > Control Panel > User Accounts > Advanced > Advanced user management , select Advanced .	The Local Users and Groups dialog box is displayed.
2	Select User > New User... from the main menu.	The New User dialog box is displayed.

3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

No	Action	Result
3	Create the TPSComServer user as shown. Use the same Password as the Administrator account for the Workgroup. 	
4	Having created the User you now must assign rights to the User.	The Local User and Groups dialog box is displayed.
5	Select TPSComServer > Properties > Member of > Add	The Select Groups dialog box is displayed.
6	Add the Administrators group for the TPSComServer user and click ok.	The administrators group gets added for the TPSComServer user.
7	Select OK .	
8	Verify that the account configured to run the TPN Server is correct.	

3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

No	Action	Result
9	Select Start > Run . Enter dcomcnfg and select OK .	The DCOMCNFG dialog box is displayed showing the list of Distributed COM applications installed on the workstation.
11	Select Console Root > Component Services > Computers > My Computer > DCOM Config > HCI_TPNServer - exe Server > right mouse click > Properties	The HCI TPNServer – exe Server Properties dialog box is displayed.
12	Select the Identity tab. Ensure that the This User option is selected, and that the user is the <Workgroup>TPSComServer. If not, select the Browse command.	The Browse for Users dialog is displayed.
	ATTENTION When checking this entry, verify that the Domain name is not that of any domain the workstation might have previously been a member of.	
13	Select the TPSComServer user from the list of users displayed. Select the Add command.	The TPSComServer user name is displayed in the Add Name area.
14	Select OK .	The TPSComServer user should now be displayed as the user that will be used to run the HCI TPN Server.
15	Select OK until the application completes.	

3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus

Upgrading TPS Builder on a GUS Node that is to remain part of a TPS Domain



ATTENTION

Refer to the *System Management Configuration Guide (EX-08)* and *TPS System Implementation Guide (TP08X)* for information relating to Windows 2003/XP and TPS Domain configuration.

This section describes requirements when there is an existing version of TPS Builder installed on a GUS Node that is part of a TPS domain, and will remain as part of a domain.

If you are installing TPS Builder on a GUS Node that is currently part of a TPS Domain, and you intend to continue to run with a TPS Domain and DC then you must perform the following before installing TPS Builder R410 or later.

- Remove the server currently associated with this GUS from the TPS domain. (see “Special Considerations when the GUS Node is not part of a TPS Domain”).
- Install HCI Client Connectivity application (see “Installing HCI Client Connectivity application”).
- Uninstall any existing version of TPS Builder (see “Uninstall TPS Builder”).

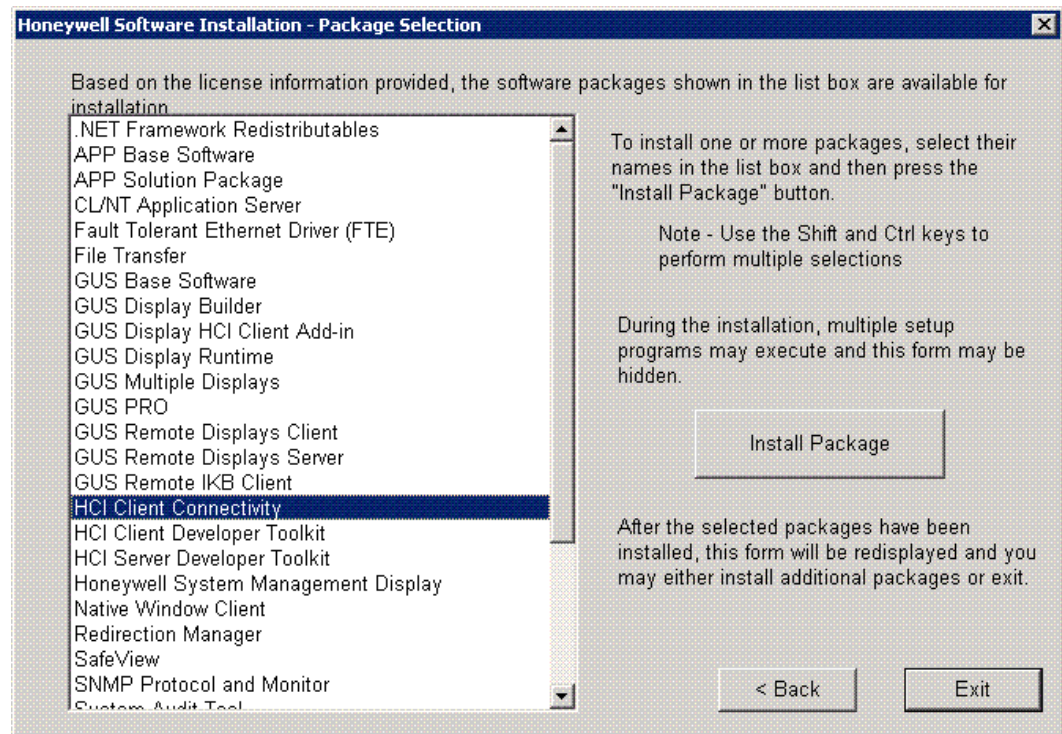
Installing HCI Client Connectivity application



ATTENTION

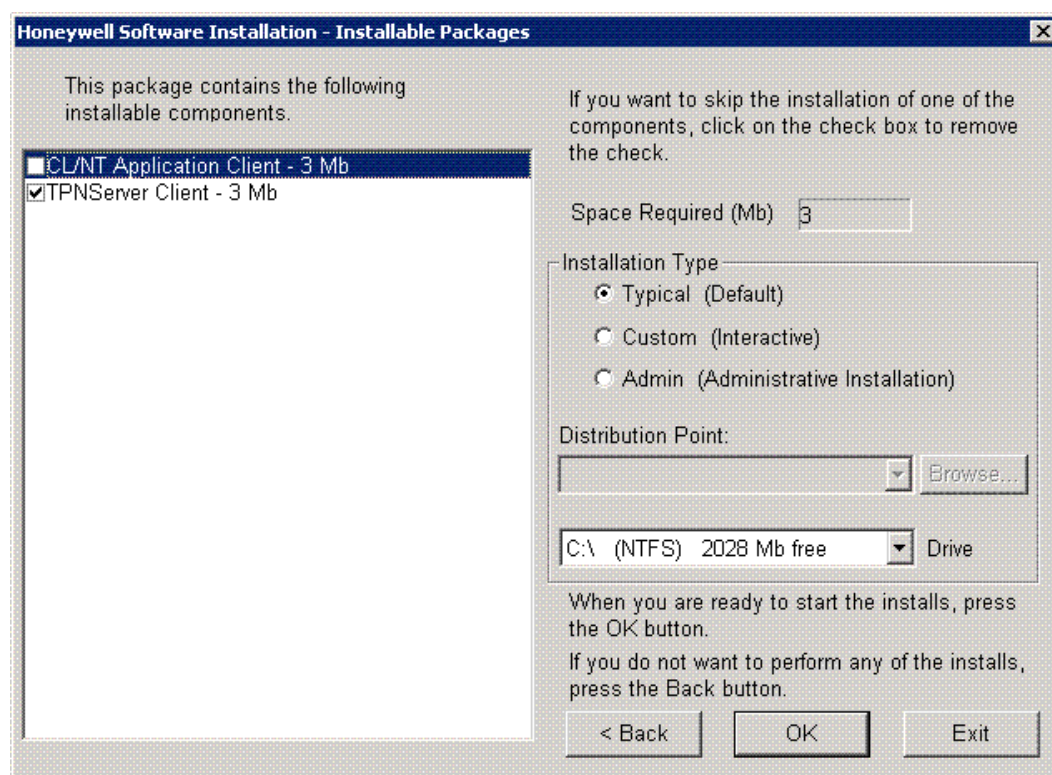
Refer to the *System Management Configuration Guide (EX-08)* and *TPS System Implementation Guide (TP08W)* for information relating to Windows 2003/XP and TPS Domain configuration.

When the existing TPN Server is uninstalled, the Server Configuration utilities are also uninstalled. If you intend to still continue to run in a TPS Domain the utilities must be re-installed by installing the HCI Client Connectivity application.



3. Software Installation

3.3. Installing TPS Builder on a GUS Node for Fieldbus / Non-Fieldbus



Configuring the TPN Server in a TPS Domain

If the GUS Node is part of a TPS Domain you must configure and replicate a TPN Server in the domain as follows:

Server Component Name:	<ComputerName>_TPSBServer (if default accepted)
BasePROGID:	Hci.TPNServer
Auto Start:	No
HCI PersistentFileName:	C:\HWIAC\Checkpoints\TPSBServer.hci
Default Access Level:	POINTBUILDER
Default Server Priority Level:	1

Upgrading TPS Builder on a GUS Node that will be removed from the TPS domain

This section describes requirements when there is an existing version of TPS Builder installed on a GUS Node that is part of a TPS domain, but you intend to remove it from the domain and run without a DC.

You must perform the following before installing TPS Builder R410.

- Uninstall any existing version of TPS Builder (see “Uninstall TPS Builder”).

3. Software Installation

3.4. SQL Server 2005 Installation

3.4 SQL Server 2005 Installation

What if SQL Server is already installed?



ATTENTION

If SQL Server 2000 Personal Edition or SQL Server 2000 Standard Edition is installed on your computer uninstall the same.

If SQL Server 2005 Standard Edition is installed and running in your network or on your personal computer for purposes other than TPS Builder, you can skip the following steps, and add a TPS Builder database to SQL Server. To accomplish this, follow these steps:

Step	Action
1	If needed, refer to SQL Server instructions for adding a database.
2	Name the database "TPSB."
3	Create a TPSB user group as explained in "Establish network user group" in this topic.

Where to install SQL Server

To review, the possibilities are as follows:

- **DC:** If you are in a TPS control network, it is highly recommended that you install SQL Server on the Domain Controller (DC) in your domain. Although not recommended, you may install SQL Server on the Global User Station on which you will install TPS Builder (and Device Save/Restore, if you install this option).



CAUTION

If you install SQL Server on a GUS in a control network, do *not* run your control process and TPS Builder at the same time. This is critical to avoid problems.

- **Other:** If you are not in a TPS network, you can install SQL Server on a network server, or an individual computer or laptop.

Review the requirements explained in "Hard Drive/Server Space Requirements" in the Installation Planning section.

SQL Server Software

The SQL Server 2005 Standard Edition software is on the TPS Builder DVD.

SQL Server installation preparation

Complete these steps before you install SQL Server 2005 Standard Edition.



ATTENTION

You need Windows Server 2000 (SP4), Windows XP edition or Windows Server 2003 for successful installation and operation. The Windows XP edition is recommended.

Step	Action
1	If you have not completed your planning, do so. Refer to "Installation Planning."
2	Navigate to the DVD drive and double-click cdbrowse.exe.
3	For network installations (TPS or other), obtain a list of administrators, engineers, and others who will be using TPS Builder or maintaining its environment. You will need their domain and logon names.

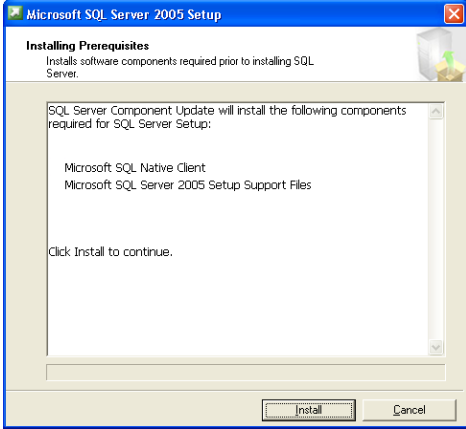
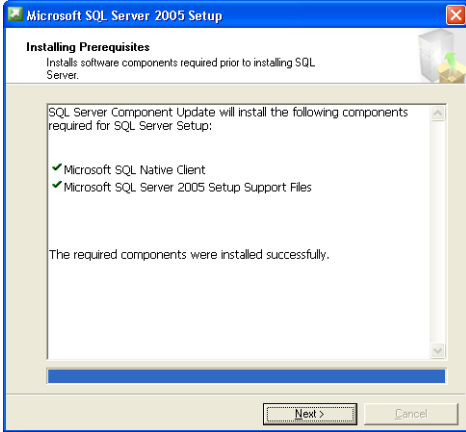
Installation procedure

Follow these steps to begin installing SQL Server from the TPS Builder DVD.

Step	Action	Results/Comments
1	If you have not completed preparation, refer to the previous subsection, "SQL Server installation preparation."	
2	Quit all Windows programs running on the system on which SQL Server is to be installed.	Ensure you also disable programs that are running automatically, such as a virus program.
3	Insert the DVD into the DVD drive and navigate to the SQL Server Folder in the root directory. (\Sql2005\cd1\setup.exe)	
4	Double-click setup.exe.	The Software License Agreement is displayed.

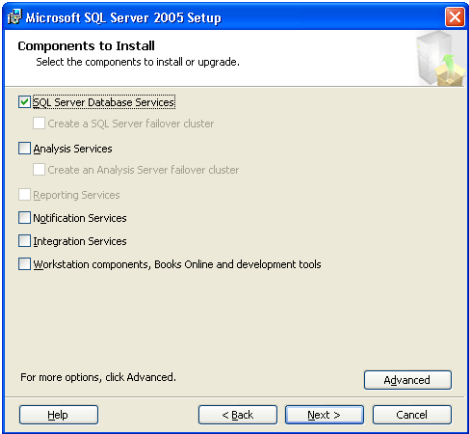
3. Software Installation

3.4. SQL Server 2005 Installation

Step	Action	Results/Comments
5	Read the license agreement, select “I accept the licensing terms and conditions,” and then click Next .	The Installing Prerequisites dialog box appears. 
6	Click Install to install the software components prior to installing SQL server.	When the installation is complete, “The required components were installed successfully” message is displayed. 
7	Click Next to continue installing the SQL server.	The Microsoft SQL server installation wizard appears.
8	Click Next .	The System Configuration Check dialog box is displayed.

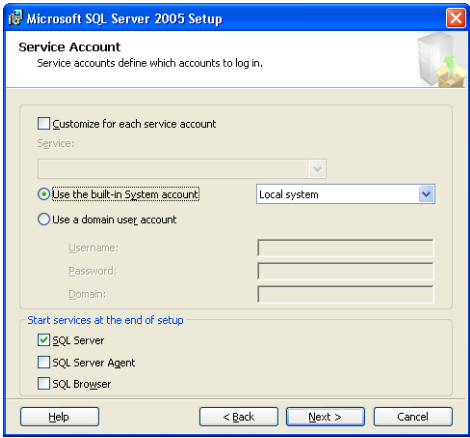
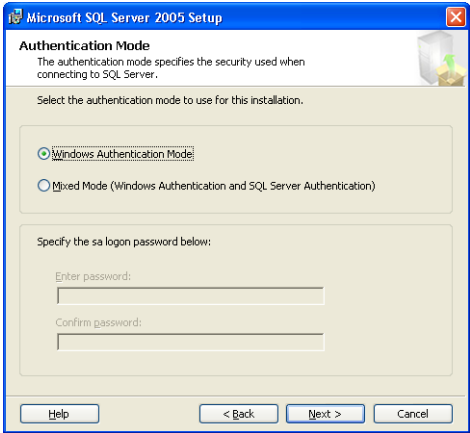
3. Software Installation

3.4. SQL Server 2005 Installation

Step	Action	Results/Comments
9	Click Next .	The Microsoft SQL Server Installation message box appears. The Registration Information dialog box appears. Note: The Product key is entered by default.
10	Click Next .	The Components to Install dialog box appears. 
11	Select SQL Server Database Services , and then click Next .	The Instance Name dialog box appears.

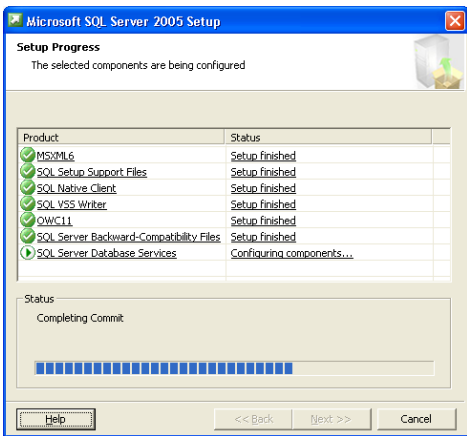
3. Software Installation

3.4. SQL Server 2005 Installation

Step	Action	Results/Comments
12	The Default instance option is selected by default. Click Next.	The Service Account dialog box appears. 
13	Select Use the built-in system account option, and then select Local system in the list that appears. Note: This option enables you to log on as an <i>Administrator</i>. Click Next.	The Authentication Mode dialog box appears. 
14	The Windows Authentication Mode option is selected by default. Click Next.	The Collation Settings dialog box appears.

3. Software Installation

3.4. SQL Server 2005 Installation

Step	Action	Results/Comments																
15	The SQL Collations option is selected by default. Click Next .	The Error and Usage Report Settings dialog box appears.																
16	Click Next . You can select any of the options available to generate the error reports.	The Ready to Install dialog box appears.																
17	Click Install .	The Setup Progress dialog box appears.  <table><thead><tr><th>Product</th><th>Status</th></tr></thead><tbody><tr><td>MS/SQL 6</td><td>Setup Finished</td></tr><tr><td>SQL Setup Support Files</td><td>Setup Finished</td></tr><tr><td>SQL Native Client</td><td>Setup Finished</td></tr><tr><td>SQL VSS Writer</td><td>Setup Finished</td></tr><tr><td>OWC11</td><td>Setup Finished</td></tr><tr><td>SQL Server Backward-Compatibility Files</td><td>Setup Finished</td></tr><tr><td>SQL Server Database Services</td><td>Configuring components...</td></tr></tbody></table> Status Completing Commit Buttons: Help, << Back, Next >>, Cancel	Product	Status	MS/SQL 6	Setup Finished	SQL Setup Support Files	Setup Finished	SQL Native Client	Setup Finished	SQL VSS Writer	Setup Finished	OWC11	Setup Finished	SQL Server Backward-Compatibility Files	Setup Finished	SQL Server Database Services	Configuring components...
Product	Status																	
MS/SQL 6	Setup Finished																	
SQL Setup Support Files	Setup Finished																	
SQL Native Client	Setup Finished																	
SQL VSS Writer	Setup Finished																	
OWC11	Setup Finished																	
SQL Server Backward-Compatibility Files	Setup Finished																	
SQL Server Database Services	Configuring components...																	
18	When the selected components are configured, click Next .	The Completing Microsoft SQL Server 2005 Setup dialog box appears.																
19	Click Finish , and then complete the installation as explained.	SQL Server installation is complete.																

3. Software Installation

3.4. SQL Server 2005 Installation

Completing the installation

Follow these steps to complete the SQL Server installation:

Step	Action	Results/Comments
20	Reboot the system.	Rebooting ensures that the SQL Server service starts.
21	Click Start> Programs>Microsoft SQL Server 2005> Configuration Tools> SQL Server Configuration Manager .	The SQL Server Configuration Manager window appears.
22	Select SQL Server 2005 Services .	The SQL Server status is displayed. Under normal circumstances the SQL Server status displayed is Running . Note: If the status is other than Running , contact Honeywell's Technical Assistance Center. Refer to "Technical Assistance" in the Introduction.
23	If this is a network installation (TPS control or other), go to "Establish network user group," which follows.	A user group must be established.

Establish network user group

For a network installation (TPS or other), the administrator must identify or create one or more specific, local user groups on the DC or server on which SQL Server is installed.

Multiple groups, such as an administrative and users group, may be established.



ATTENTION

You cannot install TPS Builder in a network unless this group(s) has been established.

For each group to be established for TPS Builder, follow these steps:

Step	Action
1	Evaluate existing groups to determine if any are appropriate for TPS Builder. - If they are, make note of them and exit this procedure. - If not, plan each group (group name, members) and follow the remaining steps to create each group.
2	On the DC or server on which SQL Server is installed, choose Start > Programs > Administrative Tools > Active Directory Users and Computers .
3	Users > Rightclick New > Group.
4	Complete the New Object Group dialog box.
5	Add the needed users to the group(s) (per original planning).
6	Click OK to establish the group.
7	Close the Active Directory Users and Computers dialog box.
8	Reboot the system to make the new group(s) visible to the network.

3. Software Installation

3.5. CodeWright Installation



ATTENTION

A consideration exists regarding the configuration of user passwords at the time user's accounts were established. If new accounts were created without the option **Password Never Expires** selected, then a problem can occur with the SQL Server service after TPS Builder is in use.

Refer to the Post-Installation Activities topic, "Correct SQL Server Password."

3.5 CodeWright Installation

Before Installation

Before installing Codewright 7.5, you must uninstall any earlier version of CodeWright you have installed.

Installation Procedure

Step	Action	Results/Comments
1	Insert the CodeWright 7.5 into the CD Drive.	The CodeWright 7.5 installer is launched and the main menu is displayed.
2	Select the "CodeWright 7.5" option.	The CodeWright 7.5 32 bit installation is initiated.
3	At the User Registration dialog enter the Serial No CW600-TRXX-11987.	
4	Continue through the dialogs, accepting the default settings as follows:	
	Installation Directory dialog	Destination folder: c:\cw32
	Installation Type dialog	Type of installation: Typical
	Initial Keymap Emulation dialog	Initial keymap emulation: CUA
	CodeWright Sync Technology dialog	Development environment to integrate with Codewright: Microsoft VC++
	Select Program Folder dialog	Program Folder: CodeWright
	The Setup continues and CodeWright 7.5 is installed. The Setup Complete dialog is displayed.	You may start CodeWright 7.5 and/or read the ReadMe files.

3.6 TPS Builder Installation

Where to install TPS Builder

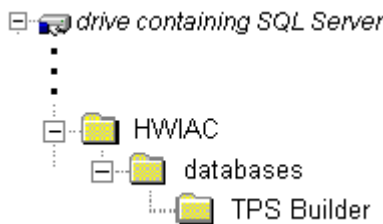
The TPS Builder application must be installed on every computer on which the program is to be installed.

CD-ROM containing TPS Builder

The **TPS Builder** CD-ROM contains all files for program installation.

TPS Builder installation preparation

Complete these steps before installing TPS Builder:

Step	Action
1	Complete installation of SQL Server 2005 Standard Edition, Microsoft Redistributables if needed as explained earlier.
2	For network installations, make sure the network user group has been established as explained in “Establish network user group” in “SQL Server 2005 Standard Edition Installation.”
3	On the system on which SQL Server is installed (DC, server, or computer), create an HWIAC folder with the following contents:  TPS Builder database files will be kept in this TPS Builder folder.
4	Make the TPS Builder folder shared: • Right-click the folder and choose Sharing. • Click Shared As. The folder name is added. Enter comments if desired. • Click OK.

3. Software Installation

3.6. TPS Builder Installation

Step	Action
5	For network installations, on the system on which TPS Builder will be installed, map a drive to the TPS Builder folder. For example,  TPS Builder on '<i>servername</i>' (X:) where <i>servername</i> is the name of the DC or server. X: is the drive letter to which you mapped the folder.
6	Network installations: On the workstation on which you are installing TPS Builder, use Windows to map a drive to the TPS Builder folder you just created. Make sure the Reconnect at Logon option is selected.
7	If the system on which you are installing TPS Builder contains a previous copy of TPS Builder, do one of the following: • If you are migrating from a previous release of TPS Builder, copy the database to a safe location and then uninstall TPS Builder as explained in "Uninstall TPS Builder." Follow the migration instructions in the Software Change Notice to migrate the database. • If you have an R140 version of TPS Builder on your system that you want to keep, continue with this installation. • If you have any version of TPS Builder that you do not want to keep, uninstall TPS Builder as explained in "Uninstall TPS Builder."
8	Locate the TPS Builder installation CD-ROM.

TPS Builder installation consists of three procedures. They are:

- Beginning the Installation
- TPS Builder-specific Installation and
- Completing TPS Builder Installation

All three procedures should be completed in a sequence. These procedures are explained in detail, in the following section.

Beginning the installation

Follow these steps to begin installing TPS Builder:

Step	Action	Results/Comments
1	If you have not completed preparation, refer to the previous subsection, "TPS Builder installation preparation."	Preparation is essential for successful installation.
2	Quit all Windows programs running on the system on which TPS Builder is to be installed.	Make sure you also disable programs that are running automatically, such as a virus program.
3	If the CD-ROM is • already inserted in your CD-ROM drive, then go to step 6. • not inserted, go to step 4.	
4	Insert the CD-ROM in the CD-ROM drive.	The Installation Welcome window appears.
5	Close the Welcome window by selecting the Exit button and then clicking Yes in response to the confirmation message.	
6	Navigate to the Builder\Packages folder on the CD-ROM and double-click Insttps.exe .	TPS Builder installation begins. Important! You may see other .exe files. Make sure you locate Insttps.exe , or the installation will not succeed.
7	Read the Welcome screen and click Next .	The Software License Agreement appears.
8	Read the entire Honeywell end-user license agreement. Click Next to agree to the license and continue.	A caution regarding third party software appears.
9	If you did not read the third-party software information in the <i>Software Change Notice</i> yet, do so now. Click Next .	

3. Software Installation

3.6. TPS Builder Installation

Step	Action	Results/Comments
10	User Information: Enter • User name and company, and • License and authorization numbers as shown on the license agreement provided with TPS Builder. When finished, click Next to go to the Package Selection screen.	Note the following: • If this dialog box has been used on this system before, the information will be filled in. In that case, check the information and correct as necessary. • If you make a mistake in the authorization entries, an error message displays. Click OK and try again. • After three unsuccessful attempts, an error message displays, informing you that software cannot be installed. Click OK to terminate the installation procedure. If this happens, contact Honeywell for assistance. Refer to “ • Technical Assistance“ in the Introduction.
11	Package Selection: Select TPS Builder and other packages that you want to install at this time. Use Ctrl+click to choose multiple packages.	A list of licensed packages displays.
12	When all desired packages are selected, click Install Package and continue with the next steps.	
The following appears only for Device Save/Restore and Redistributables:		
13	Installable Packages: If you want to install the packages shown in this list, leave the corresponding box check for the installable components.	
14	Installation Type is pre-selected as Default with a default drive location.	
15	When you are ready to start the installation, click OK .	The second phase of TPS Builder installation begins.

TPS Builder-specific installation

Follow these steps in the **TPS Builder Setup** dialog box to continue TPS Builder installation.

Step	Action	Results/Comments
16	Select Builder Components: Choose Application Files . If you want to install CL Editor at this time, choose that option, also. When desired options are selected, click Next .	
17	Select Destination Location: Accept the default path or click Change to choose another path where the TPS application files are to reside. When finished, click Next .	Important! For network installations, the application files reside on a user's system, not on the PDA or server.
18	Select Program Folder: Accept the default name for TPS Builder or enter a new folder name or choose from the Existing Folder list. When finished, click Next .	For manageability, limit this name to 20 or fewer characters.
19	TPS Builder Database Usage: Read the information about choices for the location of your TPS Builder database files. When finished, click Next .	Important! Your selection must be based on where you have already installed SQL Server and created the HWIAC\databases\TPS Builder folder. If you have not completed this step, follow the instructions in "TPS Builder installation preparation" before proceeding.
20	Specify Database Location: Click the choice that represents the location of SQL Server and HWIAC\databases\TPS Builder folder: • Local disk: For installations on a personal computer or laptop; not a network installation. • Network drive: For installations in a TPS or other network. • Existing R400 database: For users who already have a TPS Builder R400 database in place, and want to use that database. Click Next.	Important! The Existing R400 choice applies only to R400 databases. To migrate to an earlier database, cancel this installation and refer to the <i>Software Change Notice</i> for migration instructions.

3. Software Installation

3.6. TPS Builder Installation

Step	Action	Results/Comments
21	Select Database Location: For network installations, enter the drive letter of the mapped TPS Builder folder in the format X:\, where X is the drive letter for the mapped TPS Builder folder. For local installations, enter the complete path: X:\HWIAC\databases\TPS Builder, where X is the drive (partition) on which the folder resides. Create a share at the HWIAC directory level (right-click > Properties > Sharing and Security) Click the Permission option in the Sharing tab to check if the user is assigned with "Full control, Change & Read" options. If not, check all those. If you are not certain of the location of your HWIAC\databases\TPS Builder folder, click Change to locate it. When finished, click Next.	The default entry is C:\HWIAC\databases\TPS Builder, which applies to local installations only. For network installations, you must use a drive letter; this installation does not recognize UNC (Uniform Naming Convention) NetworkMachine format.
22	Repository Name: Enter a name desired for the TPS Builder repository. When finished, click Next.	The TPS Builder repository is created within SQL Server. You choose the main portion of this name, which will be prefixed with TPSB_.
23	WinNT logins: Network installations: Enter the following information: - The name of the domain in which SQL Server and the computer reside. - The name of the NT group you created during preparation. When finished, click Next.	Important! If you did not identify or create a group(s) in your domain before reaching this point, cancel this installation and follow the instructions in "TPS Builder installation preparation." Afterwards, repeat the installation.

Step	Action	Results/Comments
	Individual computer installations: - Leave all fields blank and click Next. - When asked to confirm no logins, click Yes.	Not applicable to individual installations.
24	Confirm Setup Information: Review the selections shown on this page. - If any corrections need to be made, click Back to the page requiring changes. - When the Confirm Setup Information page is correct, click Next.	The TPS Builder application and database files are now installed. A series of progress messages displays as TPS Builder installs. The installation may take a few minutes, depending on your system/network.

Completing TPS Builder installation

Follow these steps to complete TPS Builder installation:

Step	Action
1	Click Finish to exit the Wizard.
2	To quit the installation program, click Exit in the package selection screen, and in the resulting confirmation message, click Yes .
3	Restart your computer

Installation problems

If the installation does not complete successfully, an error message will display, and all TPS Builder components will be removed from the target computer (and network locations, if applicable).

Possible reasons and corrective action for installation problems are listed in the following table. For other problems, contact the Honeywell Technical Assistance Center (TAC). Refer to “

Technical Assistance“ in the Introduction.

3. Software Installation

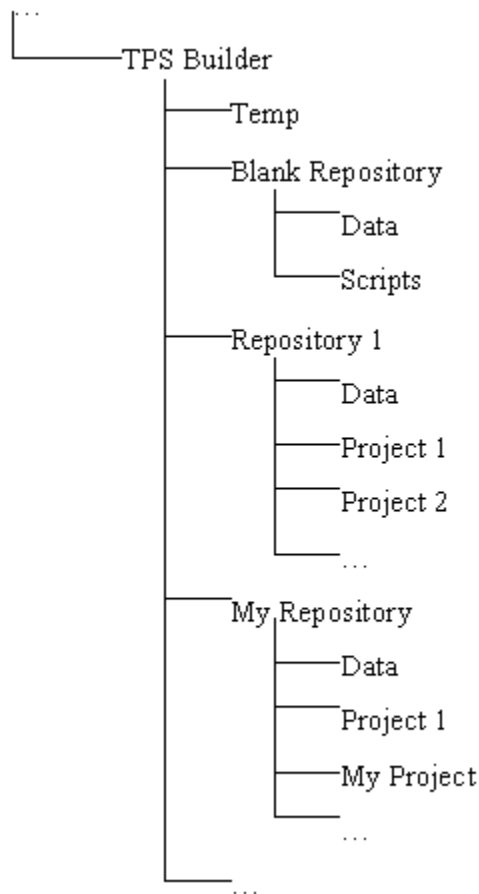
3.6. TPS Builder Installation

Problem	Action
Insufficient disk space	Increase per “Basic Installation Requirements.”
Incorrect environment	Refer to “Basic Installation Requirements” in the Installation Planning section.
TPS Builder already installed	Use the migration instructions provided in the <i>Software Change Notice</i> .
Damaged CD-ROM or CD-ROM drive	Check the CD-ROM on another computer. If the CD-ROM is damaged, call TAC. If your CD drive is damaged, replace it.

3.7 TPS Builder Folder Structure

TPS Builder repositories

The following figure shows the folder structure for TPS Builder repositories. This entire structure will be created on the TPS Builder repository, that is, where HWIAC\databases\TPS Builder was established. At the figure indicates, the TPS Builder folder is the “root” for the TPS Builder repositories.



3. Software Installation

3.7. TPS Builder Folder Structure

The contents of each repository folder are shown in the following table.

Folder	Comments
data	Contains the TPS Builder database, cs95.mdb or cs97.mdb, as well as drawing files such as S_#####, T_#####, and/or t_##### files, where ##### is a multi-digit number. As you create drawings, the drawing files are added to this folder. If you assign custom Help topics to libraries or templates, you should place the Help files in this directory.
project_name	Whenever you create a project, a folder by the same name is created. This folder contains four additional directories: • build—For your Exception Build and Entity List (.eb and .el) files • document—For documents you associate with strategies (.doc files) • linkunit—For your Link Unit (.lu) files • program—For Control Language (.cl) files you associate with strategies.
documents	If you install TPS reference documents, they are placed here.

"Program Files" contents

The \Program Files\Honeywell\TPS\Tools\Builder folder includes all application files and the following folders:

Folder	Comments
Examples	This folder holds an API script showing how to import data from an Excel spreadsheet into TPS Builder, as well as sample data in Excel. Refer to the API discussion.
Report	This folder holds report definition (.rpt) files. If you create custom reports, you would place them here, as well.

3.8 Device Save/Restore Installation

Functional description

Device Save/Restore is a separate program used to save and restore the configuration of Fieldbus devices within a TPS Network. Device Save/Restore is used in conjunction with TPS Builder.

Saving Device Configuration data using Device Save/Restore is not the same as saving TPN Configuration Data. For information on the use of Save/Restore, refer to the *Fieldbus Implementation Guide*.

Device Save/Restore is installed separately from TPS Builder.

CD-ROM containing Device Save/Restore

The Device Save/Restore files are contained on the **TPS Builder** CD-ROM.

Where to install Device Save/Restore

Before you begin your installation of Device Save/Restore, you should determine the location(s) for this installation. You may install Device Save/Restore on a server, which has multiple users, and/or it can be installed on your local workstation(s) as well. If you do install it on the same network node as SQL server, it is important to remember that your Download directory (created at the time of download) becomes the database for Device Save/Restore when you open this application.

Requirements for the Save/Restore function

Hardware requirements

The requirements for Device Save/Restore are the same as for TPS Builder.

Software requirements

Device Save/Restore must be installed in a network that has TPS Builder Release 130 or later installed.

3. Software Installation

3.8. Device Save/Restore Installation

Device Save/Restore installation procedure

Follow these steps to install Device Save/Restore:

Step	Action	Result
1	Perform steps 1 through 10 of the TPS Builder installation procedure.	
2	Installable Packages: Select Device Save/Restore Package.	
3	Click Install Package .	Device Save/Restore installation is started. The Select Destination dialog box is displayed.
4	Perform steps 13 through 15 of the TPS Builder Installation procedure.	
5	The Select Destination Location dialog box permits the selection of the directory in which the Device Save/Restore application is installed. If you want to change the default, select the Browse command and use the Choose Folder dialog box to select the destination.	The default directory is: C:\Program Files\Honeywell\TPS\Tools\DSR.
6	Click Next .	The Select Program Folder dialog box is displayed.
7	The Select Program Folder dialog box allows you to select the program folder from which the Save/Restore function is to be invoked. The default folder is Device SaveRestore . If you want to change this, enter the program folder name into the text input or select from the current list of program folders. Click Next .	The Device Save/Restore Database Usage dialog box is displayed.

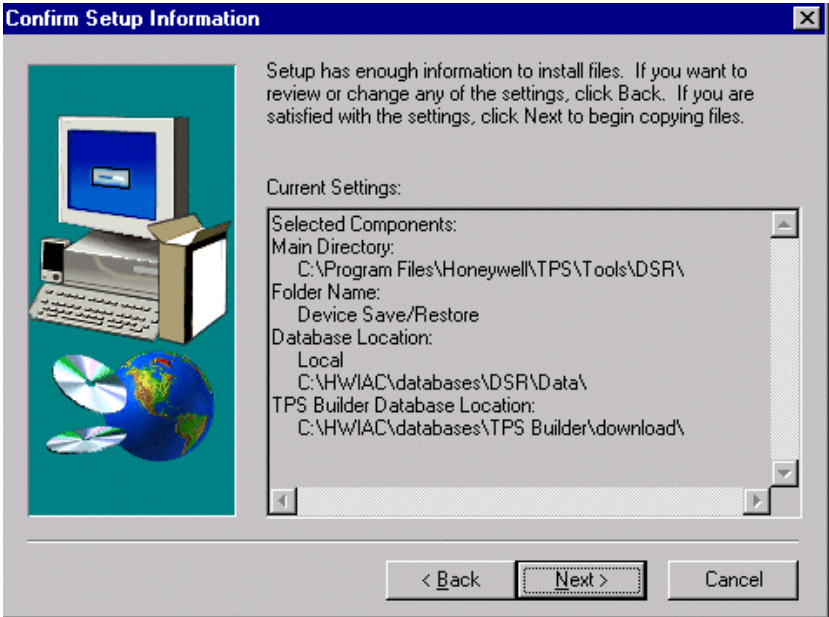
3. Software Installation

3.8. Device Save/Restore Installation

Step	Action	Result
8	Read the information presented. Options available are: Local disk – database location is on local workstation/GUS. Network drive – database location is on remote workstation. Use Existing R140 databases – specify a Save/Restore database already installed on a remote workstation/GUS. Device Save/Restore will use that database.	
9	Click Next .	The Select Database Location dialog box is displayed. You must select one of the options. If you don't select the default, you will use browse to point your installation to the desired location for installation.
10	Click Next .	This dialog box determines the directory in which the database is to be located.
11	The Select Database Location dialog box determines the location of the database, as previously explained. The default for installing local is: C:\HWIAC\databases\DSR.	
12	If you wish to change the default, click Browse and use the Choose Folder dialog box to select the destination directory.	
13	Click Next . The default is: C:\HWIAC\database\TPS\Builder\download.	The Enter TPS Builder Download Location dialog box is displayed. This allows you to enter the TPS Builder download path to which Device Save/Restore will be connected.
14	If you wish to change the default, click Browse and use the Choose Folder dialog box to select the download location.	

3. Software Installation

3.8. Device Save/Restore Installation

Step	Action	Result
15	Click Next . If you have used the defaults, they will be as shown in the following figure.	The Confirm Setup Information dialog box appears.
		
	TIP The TPS Builder Database Location entry is defaulted to C:\HWIAC\databases\TPS Builder\download. If you have installed TPS Builder in some other directory, this will not be a problem. The C:\HWIAC\databases\TPS Builder\download directory will be created automatically when you next perform a TPS Builder download.	
16	To make changes, click Back . If the selections are correct, click Next .	The Device Save/Restore application is installed. Various progress bars and dialog boxes are displayed during installation.
17	After the install is complete, a dialog box is displayed informing you that the installation has completed successfully. Click Finish .	The Install Shield Wizard Complete dialog box is displayed.

4. Post-Installation Activities

4.1 Post-Installation Additions

Types of additions

You can install TPS Builder subsystems or options after your initial installation. For example, you may have initially installed only the LCN/UCN subsystem, and decide to purchase the Data Hiway subsystem. Or, you may obtain additional Fieldbus templates.

Three types of additions can be made after your initial TPS Builder installation.

Type 1

This type of addition requires that you “unlock” a purchased subsystem or option (“package”) from your existing TPS Builder CD-ROM. You use this approach for new purchases, or if you initially purchased a package, but did not install it. Additions of this type include:

- Control Drawing-to-GUS Export Tool add-on
- Data Hiway libraries
- LCN/UCN libraries, including Fieldbus libraries

To make this type of addition, refer to “Post-Installation Addition of TPS Builder Packages” in this section.

Type 2

This type of addition requires that you import templates. Templates that can be obtained include:

- Fieldbus templates

The templates are provided in a “transfer repository” on a separate CD-ROM.

To make this type of addition, follow the instructions provided with the CD-ROM that contains the transfer repository. For instructions on importing TPS Builder objects (templates are one kind of TPS Builder object), refer to “Import TPS Builder Objects” in the Database Administration section.

4. Post-Installation Activities

4.1. Post-Installation Additions

Type 3

This addition is that of CL Editor, if it was not installed initially. To add CL Editor after you have installed TPS Builder, refer to “Post-Installation Addition of CL Editor” in this section.

User-supplied software

If you install TPS Builder before user-supplied software (such as Microsoft Excel or Crystal Reports Pro) is installed, you can still add the user-supplied software. In this case, exit TPS Builder and follow manufacturer’s instructions to install the user-supplied software.

4.2 Post-Installation Addition of TPS Builder Packages

Description

This procedure is for Type 1 additions, as described in “Post-Installation Additions” in this section.

Procedure

To make this type of addition, follow these steps:

Step	Action
1	Make a note of the confidential license and authorization numbers that are provided with your new purchase. (This number is usually provided with your license agreement.)
2	Close any applications that are running.
3	Insert the TPS Builder CD-ROM (the one provided with your latest purchase).
4	Read the following notes, then complete steps 5 – 24 of the TPS Builder “Beginning the installation.”
	Steps 5-9: Review the information if desired. Step 10: User information will be in place from the previous installation; replace the license and authorization numbers with the new ones. Step 11: Choose the TPS Builder application AND newly purchased (previously not installed) package(s). Steps 13-24: Make sure you make the same choices and entries that you did on the first installation.
5	Complete the installation as explained in the topic “Completing TPS Builder installation.”

4. Post-Installation Activities

4.3. Post-Installation Addition of CL Editor

4.3 Post-Installation Addition of CL Editor

Procedure

If you installed TPS Builder, but did not install CL Editor at the time, you can install it later by following this procedure.

Follow these steps to install CL Editor after TPS Builder has been installed:

Step	Action
1	Close any applications that are running.
2	Install Borland CodeWright. Refer to the Borland documents.
3	Insert the TPS Builder CD-ROM.
4	Read the following notes, then complete steps 5 – 13 of the TPS Builder “Beginning the installation.”
	Steps 5-9: Review the information if desired. Step 10: User information will be in place from the previous installation; you should NOT need to make any changes. Step 11: Choose TPS Builder . Step 13: Clear Application Files , select CL Editor , and then click Next .
5	Current Settings: Click Next . CL Editor installs, followed by a completion message.
6	Complete the installation as explained in the topic “Completing TPS Builder installation.”

4.4 Uninstall TPS Builder

Description

The uninstall procedure:

- removes all objects registered during TPS Builder installation;
- deletes all files installed by TPS Builder;
- leaves TPS Builder repository(ies) and document folders;
- leaves some .ocx files under Program\Honeywell\TPS\Tools\Builder.

CAUTION If you are uninstalling TPS Builder in order to re-install it or install a new version, you need to save your repository and HWIAC\databases\TPS Builder folder.

To ensure you do not lose valuable data, back-up these items to a safe location, before uninstalling TPS Builder.

Remove TPS Builder application

To remove the TPS Builder application, but not the repository:

Step	Action
1	On the desktop, click Start , then choose Programs > TPS Builder > Uninstall TPS Builder .
2	Click Yes in response to the confirmation message.
3	After the procedure completes, read any final comments in the Uninstall dialog, and then click OK to close the dialog.
4	If any components remain, such as a shortcut icon on your desktop, remove them. Other components that may remain will be located in the Honeywell folder (or other folders if specified during installation). Locate this folder, and manually delete any portions of them that you do not want. Important! If you will be re-installing or installing a new version, do not delete the HWIAC\databases\TPS Builder folder.

4. Post-Installation Activities

4.5. Stop, Start SQL Server Service

Step	Action
5	Uninstallation window will close automatically completing all the cleaning process.

Remove TPS Builder, including repository

If you do want to remove TPS Builder permanently, follow these steps:

Step	Action
1	Use the Database Administration Tool to delete the repository. Refer to "Delete TPS Builder Repository" in the Database Administration section.
2	Uninstall TPS Builder as explained in the previous procedure.

4.5 Stop, Start SQL Server Service

To stop or start the SQL Server service, follow these steps:

Step	Action
1	Quit TPS Builder.
2	Choose Start > Settings > Control Panel > Administrative Tools .
3	Choose the Services option.
4	In the Services dialog box, scroll in the Service list until you locate MSSQLServer .
5	Click MSSQLServer and click Stop or Start , as needed.
6	Close the Service dialog box.

CAUTION

Do not use TPS Builder unless the SQL Server service is in operation. Failure to do so will result in any number of TPS Builder functions crashing, and you may lose data.

4.6 Correct SQL Server Password

Problem

If you are running TPS Builder in a non-TPS network or on a personal computer/laptop, it is possible to create a mismatch between your system password and the SQL Server password. (This condition should not occur in a TPS system.)

The problem will occur when all of these conditions occur:

- When the user account was established, the Password Never Expires option was NOT selected. (This is a system or network administrator's responsibility.)
- The user's password is changed.
- The SQL Server service is interrupted (for example, you reboot the system on which SQL Server is running).

What happens

If this problem occurs, when you reboot, you will receive a message indicating that at least one service did not start. This message is referring to the SQL Server service.

Solution

To correct the problem and start the SQL Server service, follow these steps:

Step	Action
1	Choose Start > Settings > Control Panel > Administrative Tools .
2	In the Control Panel , choose Services .
3	In the Services dialog box, Service list, click SQL Server (MSSQLServer) . (Note that its status is blank.)
4	Right-click Properties .
5	In the Logon tab, Log On section, click the Local System Account option.
6	Click OK . You return to the Services dialog box; SQL Server is still selected.
7	With SQL Server selected, click Start . The SQL Server service now starts.
8	Close the Services dialog box.

4. Post-Installation Activities

4.6. Correct SQL Server Password

5. Software Configuration

5.1 Overview

Configuration considerations

When configuring TPS Builder, take into consideration whether the option being configured is one of individual preference, or should be configured in accordance with site or other requirements.

For example, consider whether or not more than one person will be working on the same projects. Also consider site standards or guidelines for such activities as creating drawings or troubleshooting.

Before configuring TPS Builder

If you are new to TPS Builder, read *TPS Builder Fundamentals*.

What can be configured

Software configuration includes establishing various TPS Builder behaviors. You can:

- Define How Projects and Libraries are Opened
- Set Up the Workspace
- Set Up Composition Views
- Define Parameter Configuration Form Format
- Define Default Size for Drawing Symbols
- Set Log Options
- Set Fieldbus Options
- Set Online Operations Data

5. Software Configuration

5.2. Define How Projects and Libraries are Opened

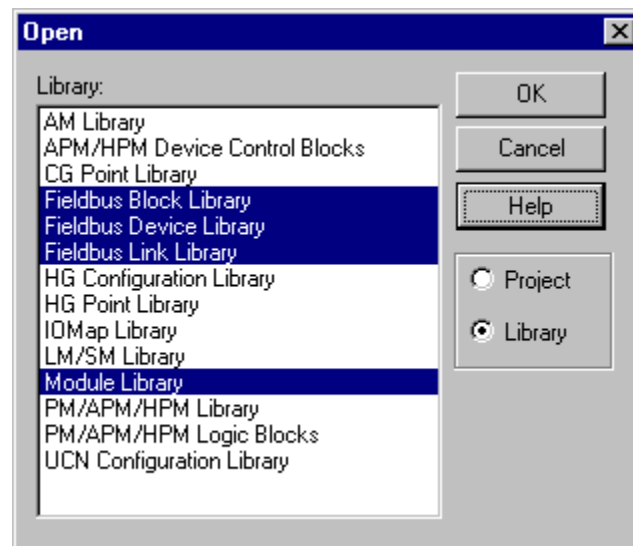
5.2 Define How Projects and Libraries are Opened

Description

You can configure TPS Builder to allow:

- a single library, or single project in one view, to be opened at a time, or
- multiple libraries, or multiple projects in one or more views, to be opened at one time.

The following figure shows multiple libraries selected at one time. When **OK** is clicked, all libraries will open in the TPS Builder workspace.



Procedure

To configure TPS Builder so that you can open multiple libraries or projects, choose **Options > Enable Open Multiple**.

If this option is selected, a check mark appears next to it:

✓ Enable Open Multiple

To restrict TPS Builder so that only one library or project can be opened at a time, clear **Options > Enable Open Multiple**. The check mark will be cleared.

5.3 Set Up the Workspace

Description

All will find the workspace arrangement that works best for them. Some suggestions are presented in the following paragraphs that may help you work most efficiently.

Libraries

In many cases, you will be working on one project, but will require several libraries. In this case, open the libraries needed and keep them open. You can size and arrange them vertically on the left side of your workspace, then minimize them if they “get in the way.” Keep them in the workspace for easier access, however.

Workspace layout

Many of you will find it easiest to work with libraries aligned on the left and projects on the right. You can size and arrange the windows as needed.

Save workspace layout

To maintain specific project and library windows in a particular arrangement for your next TPS Builder session, execute the following steps:

Step	Action
1	Open (or leave open) those windows you want available during your next session.
2	Position windows as desired.
3	Exit TPS Builder by clicking the X button in the upper right corner of the main TPS Builder window.

5.4 Set Up Composition Views

Fundamentals



REFERENCE - INTERNAL

Basic definitions and details regarding TPS Builder views are contained in the *TPS Builder Configuration Tool User's Guide*, "Composition Views."

Planning

Establishing the composition views required for your implementation activities can be considered part of TPS Builder configuration.

Sound planning will help you make the most of composition views. This includes deciding on the view in which to open new projects, and determining whether new views are needed.

Choose default view for new projects

By default, TPS Builder will open a new project in Equipment view. If you would like new projects to open in another view:

Step	Action
1	Choose Options > Set Default View .
2	In the list of existing composition views (system and custom), choose the desired view.

The view you choose will remain the default view for new projects unless you change it again.



TIP

Remember that if a new project opens in a view you do not want, you can always open it in the desired view by opening the project and selecting another view.

Create a new view

Planning

Before creating and using new views, determine what will bring you the most efficiency in dealing with projects. For example, an Application view might be created for NT applications.

When naming composition views, keep these guidelines in mind:

- The name can include any characters, including spaces (except all spaces, or spaces at the beginning or end of the name).
- Limiting the name to 15 - 20 characters will help ensure that it is completely visible in all dialog boxes in which it appears.

Procedure

To create a new composition view:

Step	Action
1	Choose Views > Composition Views .
2	In the Composition Views dialog, click Create .
3	In the resulting dialog box, enter a name of 45 characters or less. • This name can include any characters, including spaces (except all spaces, or spaces at the beginning or end of the name). • Limiting the name to 15 - 20 characters will help ensure that it is completely visible in all dialog boxes in which it appears.
4	Click OK .

5. Software Configuration

5.4. Set Up Composition Views

Change a view name

Considerations

The name of a user-defined composition view can be changed. If you do so, the name will change for all existing templates to which this view has been applied. The name will also be changed in the Open Project dialog box.



ATTENTION

If a project is open at the time you change its view name, the view name in the project title bar will not change until that project window is closed, then reopened.

Procedure

To change the name of a user-defined composition view:

Step	Action
1	Choose Views > Composition Views .
2	In the Composition Views dialog box, select the user-defined view name you want to change.
3	Click Edit .
4	In the resulting dialog box, double-click the existing entry in the Name field. If the name includes spaces, double-click and drag to select all parts of the name.
5	In the resulting dialog box, enter a name of 45 characters or less. • This name can include any characters, including spaces (except all spaces, or spaces at the beginning or end of the name). • Limiting the name to 15 - 20 characters will help ensure that it is completely visible in all dialogs in which it appears.
6	Click OK .
7	In the Composition Views dialog box, click Close .

Delete a view

When can a view be deleted?

A user-defined composition view can be deleted, as long as it has not been applied to any templates.

If the view is currently applied to a template(s), and you still want to delete it, you will need to change the view membership of the template(s) to exclude it. This can be done only if the template has not been used to create strategies.

Procedure

To delete a user-defined composition view:

Step	Action
1	Ensure that the view you want to delete is not applied to any templates. Choose Tools > Relationship Browser to check.
2	Choose Views > Composition Views .
3	In the Composition Views dialog box, select the name of the user-defined view you want to delete.
4	Click Delete .
5	In response to the confirmation message, click Yes (or No if you change your mind).
6	Click Close .

Related topics

Refer to the following topics in the *TPS Builder Configuration Tool User's Guide* as needed:

- To check a view's usage: "*Determining Object Relationships*"
- To determine whether or not a template has been used: "*Template usage*"
- To change view membership of a template: "*How to change composition view membership*"

5. Software Configuration

5.5. Define Parameter Configuration Form Format

5.5 Define Parameter Configuration Form Format

Options

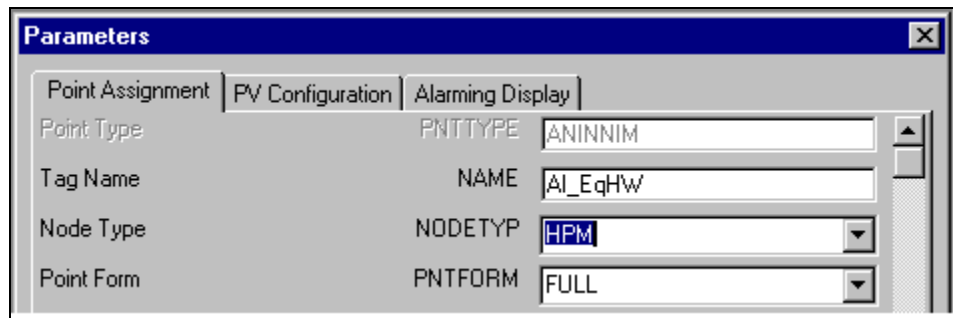
A configuration form can be viewed in one of the following layouts:

- Tabbed layout
- Alphabetized layout
- DEB-order layout

In addition, you have the option to show or hide read-only parameters. For example, if you choose to hide read-only parameters, the parameter PNTTYPE in the tabbed dialog example below would not be visible.

Tabbed layout

This layout presents logical groupings of parameters per tab (such as the Alarming Display tab for alarm-related parameters). For example:



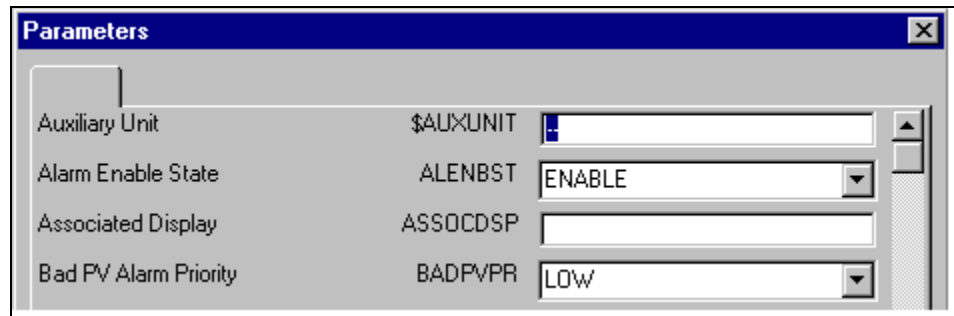
The screenshot shows a dialog box titled "Parameters" with a close button (X) in the top right corner. It features three tabs: "Point Assignment", "PV Configuration", and "Alarming Display". The "Alarming Display" tab is currently selected. Below the tabs, there are four rows of parameters, each with a label on the left and a value field on the right:

Parameter Label	Parameter Name	Value
Point Type	PNTTYPE	ANINNIM
Tag Name	NAME	AI_EqHW
Node Type	NODETYP	HPM
Point Form	PNTFORM	FULL

Each value field has a small upward arrow on its right side, indicating it might be a list box or have a dropdown menu.

Alphabetized layout

This layout is “flat” (no tabs), and all parameters are listed in alphabetical order. For example:

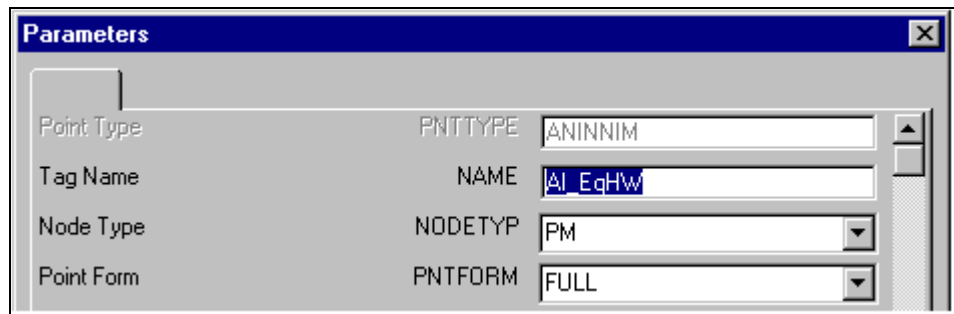


The screenshot shows a dialog box titled "Parameters" with a close button (X) in the top right corner. The dialog contains a list of parameters arranged alphabetically by name. Each parameter has a label, a variable name, and a value field. The parameters are:

Parameter Name	Variable Name	Value
Auxiliary Unit	\$AUXUNIT	
Alarm Enable State	ALENBST	ENABLE
Associated Display	ASSOCDSP	
Bad PV Alarm Priority	BADPVPR	LOW

DEB-order layout

This layout is flat, and parameters are listed in the order in which they appear in the Data Entity Builder (DEB). For example:



The screenshot shows a dialog box titled "Parameters" with a close button (X) in the top right corner. The dialog contains a list of parameters arranged in the order they appear in the Data Entity Builder (DEB). Each parameter has a label, a variable name, and a value field. The parameters are:

Parameter Name	Variable Name	Value
Point Type	PNTTYPE	ANINNIM
Tag Name	NAME	AI_EqHw
Node Type	NODETYP	PM
Point Form	PNTFORM	FULL

Read-only parameters

You can choose to view or hide read-only parameters.

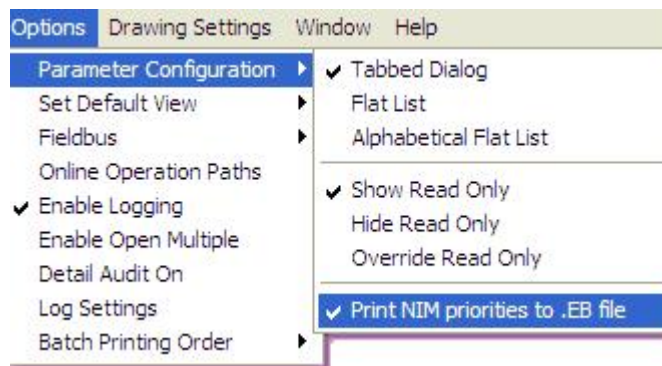
Read-only parameters are template-defined parameters. For example, if you choose to hide read-only parameters, the parameter PNTTYPE in the previous figure would not be visible at all.

5. Software Configuration

5.5. Define Parameter Configuration Form Format

Print NIM priorities to .EB file

With TPN R683/TPS R421, there is an option to display the priority parameters in the Parameter Configuration Form for the points for which the trip point parameters are set to NaN. By default, the Parameter Configuration Form does not display the priority parameters if the corresponding trip point parameters are set to NaN. To display the priority parameters in the Parameter Configuration Form, you must **choose Options > Parameter Configuration > Print NIM priorities to .EB file**.



Note: This option is independent of the **option PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN** provided in the NCF.

The following scenarios discuss about the impact on the .EB file and the PED when the option in the Network Configuration Form of TPN and the Parameter Configuration Form of TPS Builder is enabled/disabled.

Note: All the scenarios listed here assume that the alarm trip points are set to NaN.

Scenario 1:

Consider the PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN option in NCF and **the Print NIM priorities to .EB file** option is set to YES.

- Alarm parameters are visible in TPS Builder as well as in TPN.
- When the .EB file is imported to the TPS Builder, the NIM alarm priority parameters are printed in the .EB file and therefore displayed in the Parameter Configuration Form as well.
- When the .EB file is imported to TPN, the alarm priority parameters are displayed in the PED as well.

Scenario 2:

Consider the PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN option in NCF is set to NO and the **Print NIM priorities to .EB file** option is set to YES.

- When the .EB file is imported to the TPS Builder, as the NIM alarm priority parameters are not printed in the .EB file, the default values of the parameters are displayed in the Parameter Configuration Form.
- When the .EB file is imported to TPN, even though the alarm priority parameters are available in the Parameter Configuration Form, they are not displayed in the PED.

Scenario 3:

Consider the PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN option in NCF is set to YES and the **Print NIM priorities to .EB file** option is set to NO.

- When the .EB file is imported to the TPS Builder, even though the NIM alarm priority parameters are printed in the .EB file, the values of the parameters are not displayed in the Parameter Configuration Form. However, the values exist in the TPS Builder database. Subsequently, if the Parameter Configuration Form option is set to YES, the values from the database are displayed in the form.
- When the .EB file is imported to TPN, as the alarm priority parameters are not displayed in the Parameter Configuration Form, the default values of the parameters are displayed in the PED.

Scenario 4:

Consider the PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN option in NCF and the **Print NIM priorities to .EB file** option is set to NO.

- Alarm parameters are not visible in TPS Builder as well as in TPN.
- When the .EB file is imported in the TPS Builder, the NIM alarm priority parameters are not displayed in the Parameter Configuration Form.
- When the .EB file is imported in TPN, the alarm priority parameters are not displayed in the PED.

Choosing form options

The configuration form options can be changed at any time. To do so, choose **Options > Parameter Configuration**. From this submenu, choose or clear options as needed.

5. Software Configuration

5.6. Define Default Size for Drawing Symbols

CAUTION

An additional option is available: **Override Read Only**. Use this option only if instructed to by Honeywell's Technical Assistance Center (TAC).

Otherwise, you may change parameter values that should not be changed, and incorrectly configure your system.

5.6 Define Default Size for Drawing Symbols

How the default size works

You can set a drawing symbol's default size. Note that:

- This size will not affect existing drawings, nor templates that have already had any drawing properties set.
- It will be used the first time you set any drawing properties for a template.
- It will also be used the first time you open a drawing for any of the drawing's strategies created from templates without drawing properties set.

Preparation

Begin by evaluating your drawing symbol needs. You will need to experiment with different sizes to determine what is appropriate. In considering drawing symbol size, you must also consider

- the information that is to appear on the symbol (in addition to the strategy and template name, you may want to include parameter names and values),
- the font type and size that are appropriate, and
- the number of connectors for the symbol.

Also consider the total number of strategies to be included in your drawings.

With this information in mind, continue with the procedure.

Procedure

You may want to perform this procedure to define sizes for templates for different types of strategies, such as control strategies, input/out strategies, and so forth.

To establish a default size:

5. Software Configuration

5.6. Define Default Size for Drawing Symbols

Step	Action
1	Choose Drawing Settings > Default Build Symbol Size . . .
2	In the Set Default Build Symbol Size dialog box, choose a width and/or height, in pixels. Note that the initial Honeywell-provided size is 200x200 pixels.
3	Click OK .
4	Identify a template that has not had any drawing symbol properties set (for example, choose a template you have never used).
5	Create an empty strategy (or specialize a template) module to be used just for experimentation.
6	Open the drawing for this module.
7	Drag the template into the module drawing.
8	Evaluate the drawing symbol size on the screen and by printing a copy of your drawing. Change other properties on the strategy to match the needs you identified in step 1.
9	If the size is all right, then you have set the default size. Go to step 11. Otherwise, go to step 10.
10	If the size is not all right then set a different default size and drag the template into the drawing again. Repeat until the size is as desired.
11	If you will be using this size for all templates (and subsequent strategies) of a particular category, then go to step 12. Otherwise go to step 14.
12	In turn, select each template and set at least one other drawing symbol property (color, font, picture, parameters, or connectors. This action creates a drawing symbol properties file for the template, which includes the default size.
13	Repeat the entire procedure, choosing another drawing symbol size, for your other category of templates.
14	When you are finished, delete the experimental module.

5.7 Set Log Options

Description

Logs are used for diagnostic and other purposes. Setting related options allows you to control how much, and the types of, information logged. Available options are described in the following paragraphs.

Enable logging

Turn the TPS Builder logging function on or off. If off, no messages are written in any TPS Builder logs.

To enable logging, choose **Options > Enable Logging**.

When this option is selected, a check mark appears next to it: ☒

To stop all logging, choose the option again. The check mark will be cleared.

Enable detailed audit

Setting this option “ON” causes the Audit Trail log to record all changes to parameters, objects, and templates, including all the TPS Builder actions taken during scheduling.



ATTENTION

Setting this option on will quickly create a huge audit log. It is recommended that you use this option only if instructed by Honeywell TAC.

If TAC so recommends, choose **Options > Detail Audit On**.

When this option is selected, a check mark appears next to it: ☒

To stop the detailed audit, choose the option again. The check mark will be cleared.

Manage log size

You can manage the size of each log by

- setting the maximum number of entries allowed in it, and
- setting its overwrite options.

To make these settings:

Step	Action
1	Determine the size and overwrite requirements for a log of interest.
2	Choose Options > Log Settings .
3	Complete the Log Settings dialog box for the log of interest.
4	Click OK .
5	Repeat for other logs as needed.

5.8 Set Fieldbus Options

Description

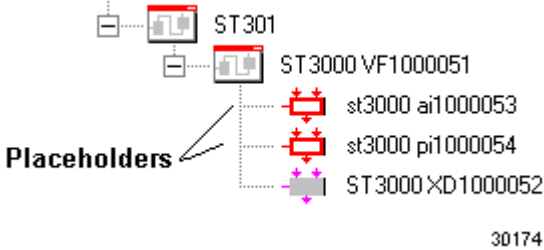
If you have installed the Fieldbus options for the LCN/UCN libraries, the following configuration options are available:

- Use of placeholders
- Error reporting
- Device matching

These options are discussed in the following paragraphs.

Use of placeholders

This option changes the composition of Fieldbus device strategies within a project as follows:

Option	Description
Placeholders On	When a Fieldbus device strategy is created from a Fieldbus device template, the function blocks that are part of the device are NOT created. A “placeholder” is used to indicate that the function blocks are present in the device. The placeholder appears as follows: 
Placeholders Off	When a Fieldbus device strategy is created from a Fieldbus device template, the function blocks that are part of the device are created as part of the device.

Use placeholders if you are creating the function blocks separately, for example, as part of control loops. To turn them on, choose **Options > Fieldbus > Placeholders Only**.

When this option is selected, a check mark appears next to it: ☒

To turn off placeholders, choose the option again. The check mark will be cleared. Placeholders already used will remain. New Fieldbus device strategies will contain function block strategies, not placeholders.

Error reporting

You can configure:

Option	Causes a Message to Display if...
Prompt on Download Errors	Download errors occur.
Prompt on Upload Errors	Upload errors occur.

Choose either or both options from the **Options > Fieldbus** submenu.

When an option is selected, a check mark appears next to it: ☒

To turn off either option, choose it again. The check mark will be cleared.

Device matching

When downloading, if a device I.D. does not match, you will “push” the TPS Builder Device physical device tag and node address to the FBIOP unless you turn off this option.

This option is on the **Options > Fieldbus** submenu.

When an option is selected, a check mark appears next to it: ☒

To turn off device matching, choose **Options > Fieldbus > Device Matching**. The check mark will be cleared.

5.9 Set Online Operations Data

Description

In order to download configuration data from TPS Builder to nodes and points on the TPS Network (TPN), online operations data must be specified for TPS Builder. You can set this path information once for TPS Builder, and it will be used for all download activities.

Operations data includes:

- **TPS Network Path:** The complete path to the History Module (HM) directory to which you will download configuration data. This path includes the drive letter designated for the HM, and the HM volume and directory names.
- **EC File:** A five-character name for the Execute Command (EC) file TPS Builder will create as part of the download process. Do not include an extension (.ec), as TPS Builder will do this.
- **EB File Path:** The complete path for the TPS Builder project-related folder in which you want Exception Build (EB) files created. (TPS Builder creates the EB files for your project first, and then loads them to the HM in a separate step).
- **Entity Load Override option:** If you are downloading data for existing entities, this option must be selected to allow the existing data to be overridden. If for some reason, you cannot override the existing data, then do not use the TPS Builder download function. (The download will fail if this option is not selected.)

Preparation

Before setting the online operations data in TPS Builder:

Step	Action
1	Identify the History Module (HM) volume and directory into which configuration data files will be downloaded.
2	On your GUS node, identify the drive designated for the HM. If the HM has not yet been mapped as a network drive on the GUS, then map it now.
3	Determine a name for the EC file that will be created. Use a name of five characters. Do <i>not</i> include the extension, ".ec." (TPS Builder will do this.)
4	Identify the TPS Builder project-related folder in which you want Exception Build (EB) files created. (TPS Builder creates the EB files for your project first, and then loads them to the HM in a separate step). The default location for these files is <i>Projectx</i> \build, where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.) You may choose another location if desired, however.
5	Note whether there is any reason you would <i>not</i> override existing entity data with the data of your current download.



ATTENTION

A TPS Builder download will fail unless you choose to override existing entity data. If you cannot override existing data, then do not use the TPS Builder download function.

Procedure

To set online operations data for TPS Builder:

Step	Action
1	Complete preparation discussed above.
2	On the Options menu, choose Online Operations Paths .
3	Complete the Default Paths for Online Operations dialog box.
4	Click OK .

These settings will be used for any download of configuration data to the TPN.

5. Software Configuration

5.9. Set Online Operations Data

6. Database Administration

6.1 Overview

Using this information

This section explains basic file information and also describes the database-related procedures provided by the TPS Builder Data Administration tool.

You need to understand the TPS Builder files in order to:

- Perform backups
- Set up specialized templates and/or libraries before distributing the database
- Use multiple databases for multiple projects, and
- Move the database for some reason.

The tasks explained in this section are considered database administration tasks, although who actually performs them may differ, depending on how your company is organized.

Function Summary

Database Administration functions are divided into two categories:

- General Utilities, and
- Administrator Utilities

The General Utilities include the following function:

- Switch repository

The Administrator Utilities include the following functions:

- Backup Repository
- Restore Repository
- Export Repository
- Import Repository
- Migrate Repository
- Create Repository
- Delete Repository

6. Database Administration

6.2. TPS Builder Files

Space availability

Ensure that adequate database space is available on individual computers and/or on the server where the database will reside. Refer to “Hard Drive/Server Space Requirements.”

6.2 TPS Builder Files

Honeywell-supplied database files

The TPS Builder database, as shipped from Honeywell, consists of:

- A Microsoft Access database, cs97.mdb or cs95.mdb (depending on which version of Access you have).
- A set of template configuration files, named T_####_0, where #### is a unique template identification number of one or more digits.

Once installed, these files are located in the HWIAC/databases/TPS Builder/data folder (or a different path, if you did not use the default installation).

User-created database files

Specialized templates that you create will have the same type of file names as those provided by Honeywell: T_####_0, where #### is a unique template identification number of one or more digits.

Drawing files that you create are identified with numbers similar to the following (where #### is a unique numerical identifier):

T_####_2 = Template drawings in Equipment view

T_####_3 = Template drawings in Control view

T_####_4 = Template drawings in Hardware view

S_####_2 = Strategy drawings in Equipment view

S_####_3 = Strategy drawings in Control view

S_####_4 = Strategy drawings in Hardware view

As they are created, these files are added to the HWIAC\databases\TPS Builder\data folder (or a different path, if you did not use the default installation).

Project files

For each project you create, a folder is created in the HWIAC\databases\TPS Builder folder. The project folder includes all files related to the project, such as .eb files, .cl files, .lu files, and so on.

Repositories

In this section and in the remainder of this document, the term "Repository" will be used to designate ALL of the files associated with the TPS Builder, including the SQL database file.

6.3 Backup TPS Builder Repository

Overview

You can back up any TPS Builder repository to a designated target directory on the SQL server (and restore it later from the same repository using the Restore TPS Builder Repository function).

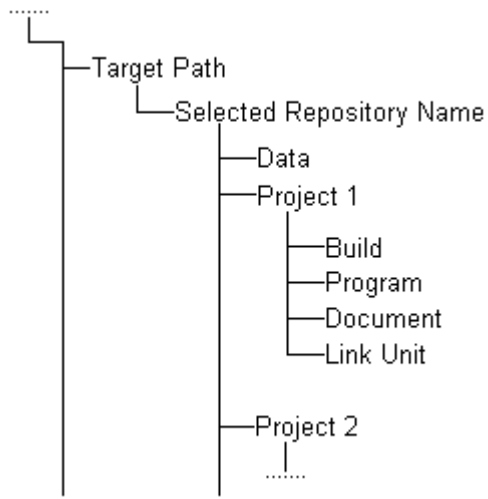


ATTENTION

Because you are backing up a repository that contains all of the TPS Builder files, you cannot do a partial backup using this function.

A TPS Builder repository includes the actual data files and the project files. The .mdf and .ldf data files are backed up to a single file with the same name and a .bak extension. The .bak file is stored under the specified target path\repository path\data directory.

The project files include t_* files, s_* files, attached documents, attached CL files, *.EB files, and *.ZT files. The project files for each project are copied directly into the target path\repository path. The adjacent diagram shows the directory structure used.



When you should back up

Ensure that a database backup procedure is established. Best practices call for

- a daily onsite backup and
- a weekly offsite backup.

Permissions

You must have administrator privileges on the SQL server to execute this function.

Procedure

To backup a TPS Builder repository, execute the following steps:

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed.
5	In the right pane, click the Backup Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears, displaying the current TPS Builder repository in the combo box.
7	To backup a different repository, locate and then select the TPS Builder repository you want to backup.
8	Click Next . Another dialog box appears.
9	Using the Browse button, locate and then select the target directory where you want to locate the backup repository.
10	Click Finish . A confirmation dialog box appears, listing the SQL server name, and the data you have input/selected in previous steps.
11	If the data is correct, click OK to complete the backup. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
12	A progress bar displaying the database backup completion percentage is displayed. If desired, click Cancel to terminate the backup procedure.
13	When the database backup is completed, a progress bar displaying the repository backup completion percentage is displayed. If desired, click Cancel to terminate the backup procedure.

6.4 Restore TPS Builder Repository

Overview

You can restore any TPS Builder repository that was previously backed-up to a directory on the SQL server using the Backup TPS Builder Repository function.



ATTENTION

Because you are restoring up a repository that contains all of the TPS Builder files, you cannot do a partial restore using this function.

Permissions

You must have administrator privileges on the SQL server to execute this function.

Procedure

To restore a TPS Builder repository, execute the following steps:

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed.
5	In the right pane, click the Restore Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears, displaying the current TPS Builder repository in the combo box.

Step	Action
7	To restore a different repository, locate and then select the TPS Builder repository you want to restore.
8	Click Next . Another dialog box appears.
9	Using the Browse button, locate and then select the data-store directory from which the selected repository will be restored.
10	Click Finish . A confirmation dialog box appears, listing the SQL server name, and the data you have selected in previous steps.
11	If the data is correct, click OK to complete the restore. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
12	A progress bar displaying the database restore completion percentage is displayed. If desired, click Cancel to terminate the restore procedure.

6.5 TPS Builder Distribution Options

Planning for distribution

Planning for distribution will help you make the most of TPS Builder. Take time to evaluate your plant's needs, existing implementation strategies, and the engineering environment to determine the best application of libraries, templates, projects, and strategies.

For large organizations in which several engineers are involved, consider the value of user libraries for optimum consistency in the creation of TPS Builder projects.

Distribution options

Regarding the sharing and transfer of TPS Builder data, you have the following options:

Method	Description
Sharing templates in order to standardize the work within an organization.	Install database on a network, giving permission to everyone whom is to use it.
Perform an initial distribution of specialized templates	With planning and preparation, specialized templates can be distributed to other TPS Builder installations.
Export libraries, templates, projects, or strategies from one TPS Builder database and import into another.	This action allows the update or modification of any TPS Builder database on an as-needed basis

6.6 Distributing Templates

Preparation

To create specialized templates and/or user libraries before distributing the database for multiple users when they first receive TPS Builder, begin with the following steps:

Step	Action
1	Install TPS Builder on the computer where the specialization will take place.
2	Familiarize yourself with procedures for specializing templates and creating user libraries.
3	Plan and create needed specialized templates and/or user libraries.
4	Make a backup copy of the specialized database and all associated files by copying them from the HWIAC\databases\TPS Builder\data folder (or a different path, if you did not use the default installation) to an appropriate backup location.
5	Prepare guidelines on using the specialized templates (and any other guidelines that you want to establish).

Distributing the database

After the specialized templates have been created, distribute them as explained in these steps:

Step	Action
1	Place a copy of the specialized data folder in a location that can be accessed by the sites where TPS Builder will be installed, or make a copy(ies) to accompany the TPS Builder CD-ROM.
2	Install TPS Builder at the additional sites (per the licensing agreement). Ensure that all sites install the same options (libraries and/or add-on). For the initial CD-ROM installation, choose the cs97.mdb or cs95.mdb database as required.
3	Once TPS Builder is installed, copy the specialized data folder to the location specified during installation for the database. The default is HWIAC\Databases\TPS Builder (overwrite the originally installed folder).
4	Distribute the guidelines created during preparation.

By following these steps, all users will have the same specialized templates and/or user libraries.

6.7 Export TPS Builder Repository

Overview

You can export any TPS Builder repository to an external file. This is accomplished by first creating an MS Access database (.mdb file) from the SQLServer database, copying all the files in the project folders, and all the drawing files, attached documents, attached CL files, etc. and then compressing all this data into a single ZIP file.

Permissions

You must have administrator privileges on the SQL server to execute this function.

Procedure

To export a TPS Builder repository, execute the following steps:

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed.
5	In the right pane, click the Export Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears.
7	In the combo box, locate and then select the name of the repository you want to export.
8	Click Next . Another dialog box appears.
9	Using the Browse button (...), locate and then select the directory (folder) where you want Zip file containing the export data to be placed.
10	Enter the name of the compressed (Zip) file.
11	Click Finish . A confirmation dialog box appears, listing the data you have input/selected in the previous steps.
12	If the data is correct, click OK to complete the export. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
13	A progress bar displaying the database export completion percentage is displayed. If desired, click Cancel to terminate the export procedure.

6.8 Export TPS Builder Objects

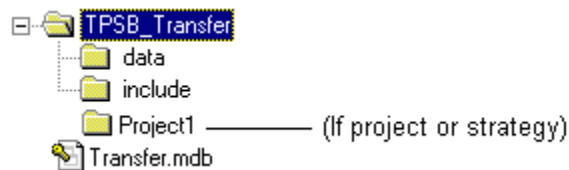
Why exporting takes place

If you have created and/or configured TPS Builder “objects” (libraries, templates, projects, and/or strategies, plus associated files) that you want to share with someone using a different TPS Builder repository, you can export the objects. Then, the templates can be imported by the other TPS Builder.

For example, if you at Site A, and want to share templates that you have specialized with someone at Site B, you need to export the templates and transfer them to Site B, where the templates could be imported.

Transfer repository

The folders and files prepared for the export are collectively referred to as the “transfer repository.” This repository appears as follows:



The transfer repository files and folders are as follows:

File/Folder	Contents/Role
data folder	All drawing files (S_ and T_ files), plus any display files attached to a template (.gus files) for the object being exported.
include folder	If you want to include files in the export, you do so by manually adding them to this folder.
<i>Project</i> folder(s)	The entire folder structure of selected projects or strategies being exported. Also, - For projects, all files related to the project (such as attached CL programs or documents) are included here. - For strategies, only those files attached to the strategy being exported are included.
Transfer.mdb	The transfer database itself.

What is exported?

The following TPS Builder objects can be exported:

- Library
- Project
- Template block
- Strategy block
- Template module
- Template module

Along with these objects, “dependent objects” are exported, as listed in the following table.

Object	Dependent Objects
Library	• The library itself. • All templates within the library (including all dependent objects as described later in this table), if you so choose (individual templates can be selected for export if desired). • Help file, if one has been attached to the library and is located in the default location of HWIAC/databases/TPS Builder/data (or equivalent location, if you did not use the default installation).
Template block (which may or may not be part of a module)	• The template with all of its views (Equipment, Hardware, and so forth). • The template from which this template was specialized (the “parent” template (goes up to the system parent)). • Drawing files (T_ files) (for a block, this file contains drawing property information) • If they have been defined for the template, – Custom parameter definitions – Custom parameter enumeration types and values – Custom parameter groups • Help file, if one has been attached to the template and is located in the default location of HWIAC/databases/TPS Builder/data (or equivalent location, if you did not use the default installation).

Object	Dependent Objects
Template module	- The module itself, including all of the objects listed in this table for a template block. - All of the blocks and/or modules contained within the module, and all of the templates from which they were specialized. - OLE object(s) inserted into the module drawing by any method. - Associated GUS display(s) that have been created using the GUS Display Conversion Tool. - Drawing files (T_ files) (for modules, this file contains drawing property information as well as information for the drawing itself).
Project	- The project with all of its views. - All strategies within the project, with all of their views, if you so choose (individual strategies can be selected for export if desired). - Objects associated with the strategies, as explained for strategies.
Strategy block (which may or may not be part of a module)	- The strategy with all of its views. - The template from which the strategy was created and all associated objects as defined for template blocks. - The project containing the strategy. - Files that have been associated with the strategy, including - CL program(s), located in the default location of HWIAC\databases\TPS Builder\Projectx\program folder, where Projectx is the name of your project. - Document(s), located in the default location of HWIAC\databases\TPS Builder\Projectx\document folder, where Projectx is the name of your project. - Drawing files (S_ files) (for a block, this file contains drawing property information). - Fieldbus strategy has few more restrictions.
Strategy module	- The strategy module itself, including all of the objects listed in this table for a strategy block. - All of the strategy blocks and/or modules contained within the module and all of the templates from which they were created. - OLE object(s) inserted into the module drawing by any method. - Drawing files (S_ files) (for modules, this file contains drawing property information as well as information for the drawing itself).

What is not exported?

TPS Builder will not necessarily export all files associated with your TPS Builder objects. For example, include files in CL and all the files in a VC++ project will not be exported.

Complete the steps in “Export objects procedure” in this topic to ensure all desired files are included.

Plan the export

Planning is the key to a successful export. Considerations include:

For the...	Consider this...
Object(s)	In TPS Builder, use appropriate procedures to identify and/or create the object(s) to be exported. • Refer to “What is exported?” preceding this topic for a complete list of what will be exported, per object type. • Ensure that all aspects of the object(s) are configured as desired before exporting.
Folder	Identify the folder into which you want the exported files to be placed. This can be any folder available on your system. (If desired, you can create folders as you export.) CAUTION If you attempt to export into a folder that has already been used for the export of TPS Builder objects, you will be cautioned that the folder will be overwritten. Do not use such a folder unless it is acceptable to overwrite previously exported objects.
Drive	Ensure the drive on which the exported files will be created has sufficient free disk space. The minimum space required for even a small export is approximately 11 MB.
Log	Decide whether you will overwrite the Export/Import log. This log is accessed by choosing Reports > Logs > Export/Import. If you do not overwrite the log, then all new export activity will be added to the log.
Time	Consider time requirements. Depending on your system, it may take several minutes to transfer a small library or project. Ensure you have sufficient time available for the export.



PLANNING TIPS

- If the templates to be exported are in separate libraries, consider moving them to a single user library so that they can all be exported at the same time (by exporting the library).
- The Export/Import log can become very large. You may find that it is easiest to use the log if you overwrite the log with each export or import.

Export objects procedure

After planning for the export is completed, follow these steps to export TPS Builder objects (libraries, templates, projects, and/or strategies, plus associated files):

Step	Action
Beginning Steps	
1	Open the library or project that is to be exported, or that contains the templates or strategies to be exported.
2	Select the object to be exported. Tips: • For a library or project, ensure no individual templates or strategies (respectively) are selected. Click directly on the library or project name. • For multiple templates or strategies within a library or a project, you can select multiple templates or strategies by using Shift+Click or Ctrl+Click as needed.
3	Choose File > Export .
Export (Step 1)	
4	In the Export (Step 1) dialog box, Folders/Drives list, navigate to the folder that is to receive the exported files. Double-click drives/folders to open them in this list. (Note: The Contents list can only be viewed, not used for navigation.) If the exact folder does not exist, you may rename an existing folder to better suit your needs or create a new folder. If you need help with either of these procedures, refer to “ Working with folders in Export (Step 1)” in this topic.
5	With the desired folder selected in the Folders/Drives list, click Next .

6. Database Administration

6.8. Export TPS Builder Objects

Step	Action
Export (Step 2)	
6	In the Export (Step 2) dialog box, evaluate the Objects selected list if needed. To review the basic facts about each entity, click it and review the Source Repository information. For instructions on navigating in the Source Repository section of this dialog box, go to "Evaluating Export and Import Details" in this section. Otherwise, go to the next step.
7	If you want to overwrite the Export/Import Log, make sure that option is selected.
8	Click Next.
9	If asked to confirm the overwrite of the Export/Import log, answer Yes or No as desired.
Export (Step 3) - Summary	
10	In the Export (Step 3) - Summary Dialog Box, review the transfer repository path and the objects selected for export. If the information is not correct, click Back until you reach the step that needs to be changed.
11	Once the information is correct, click Finish. A progress indicator shows the status of the export.
12	When the export is completed, an acknowledgement message appears. Click OK.
13	If you want to include files not included in the export, you can now add them to the "include" folder in the transfer repository. If required, refer to "Transfer repository" in this document.

Cancel an export

If you have to cancel an export while it is in progress, follow these steps:

Step	Action
1	In the progress indicator, click Cancel .
2	In response to the confirmation message, click Yes .
3	In response to the cancellation notice, click OK .

Export problems

In some cases, an export or cancellation thereof may fail. In this case, you should delete the folders created by the export, in order to increase your disk space. To do so, follow these steps:

Step	Action
1	Quit TPS Builder.
2	Using Explorer or My Computer, navigate to the HWIAC/databases/TPS Builder/ folder (or equivalent location, if you did not use the default installation).
3	Locate the Temp folder, and in it, the Transfer folder.
4	Delete the Transfer (or Transferi) folder, where i is a number.
5	Navigate to the folder in which you directed TPS Builder to place the exported files. (Refer to step 4 of "Export objects procedure" in this section.)
6	Delete the TPSB_Transfer folder (and its contents).

Working with folders in Export (Step 1)

If you need to rename or create a folder after you have chosen **File > Export**, refer to the following procedures.

Renaming a folder

To rename an existing folder, execute the following steps:

Step	Action
1	In the Export Wizard, Folders/Drives list, navigate to the folder that is to be renamed.
2	Click the name in the Folders/Drives list, pause briefly, and then click again.
3	Modify the name as desired.
4	Click outside the name. The folder is renamed and selected.
5	Continue with the export procedure. Note: The folder modifications are made on your system and will remain even if you cancel the export at this time.

Creating a new folder

To create a new folder, execute the following steps:

Step	Action
1	In the Export Wizard, Folders/Drives list, navigate to the folder in which the new folder is to reside.
2	Click New Folder .
3	Type the new folder's name.
4	Click outside of the name. The folder is selected.
5	Continue with the export procedure. Note: The folder modifications are made on your system even if you cancel the export at this time.

6.9 Import TPS Builder Repository

Overview

You can import any TPS Builder repository that has been exported to an external ZIP file using the Export Repository function into a new TPS Builder repository.

Permissions

You must have administrator privileges on the SQL server to execute this function.

Procedure

To import a TPS Builder repository, execute the following steps:

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed.
5	In the right pane, click the Import Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears.
7	Now, you can either • use the Browse button (...) to locate and then select the external Zip file for the exported repository you want to import, or • input the absolute path of the zipped file directly in the data input port.
8	Click Next . Another dialog box appears, containing a data input port for the name of the new repository, with the first 5 characters of the name set to TPSB_.
9	Enter the remainder of the new repository name and click Next . Another dialog box appears.
10	Enter the Domain Name and NT Group for the login to be added.

6. Database Administration

6.9. Import TPS Builder Repository

Step	Action
11	Click Add . The login data just entered will appear in the List of logins to be added.
12	Click Finish . A confirmation dialog box appears, listing the data you have input/selected in the previous steps.
13	If the data is correct, click OK to complete the export. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
14	A progress bar displaying the database import completion percentage is displayed. If desired, click Cancel to terminate the import procedure.

6.10 Import TPS Builder Objects

Description

After a transfer repository is created, objects within it can be imported into another TPS Builder repository.



ATTENTION

The version of the TPS Builder into which objects will be imported must be the same as the version of TPS Builder that created the transfer repository.

Transfer repository source

A transfer repository can be created by exporting objects from TPS Builder. For information on creating a transfer repository, refer to “Export TPS Builder Objects” in this section.

Special-order templates can also be imported. These templates are provided by Honeywell organizations as follows:

- Interoperability Lab: Specific Fieldbus device templates

For information on ordering these templates, contact your Honeywell sales representative.

What is imported?

The results of an import, including what happens if objects of the same name, type and location as those already in the target repository, are explained in the following paragraphs:

- Results of template import
- Results of strategy import
- Results of library import
- Results of project import

Terminology used

For the purposes of discussing import activities, the following terms are used:

- The TPS Builder repository into which you are importing is called the “target repository.”
- The object(s) you are importing is called the source or the transfer object, depending on the context.
- A “parent” template is the template from which another template has been specialized.
- A “default” library is
 - a library in which a template already resides (if this library exists in the target repository), or
 - a library in which the template’s parent resides.

Results of template import

Templates can be imported individually, and they can also be imported as the result of importing libraries, strategies, and projects.

When a template is imported, associated files, such as drawing files, template Help a file inserted into a template drawing, are also imported. In terms of the template itself, results of an import are explained in the following table.



ATTENTION

The TPS Builder repository into which objects are being imported must have the correct subsystems installed.

For example, if you want to import Data Hiway objects, the Data Hiway subsystem must have been purchased and the appropriate authorization number used during installation. (The authorization number unlocks libraries associated with the subsystems purchased.)

When the template is...	Import Selection	Results of Import	
		Object is in Target Repository	Object is Not in Target Repository
itself being imported	Place in default library	- Existing template is replaced by imported template. - If imported template's parent differs from that of the existing template, the parent is reset. - Library location is unchanged.	- Template is imported, and placed directly under its parent template. - Imported template's parent is created, if it does not exist in target repository.
	Place in existing user library	- Existing template is replaced by imported template. - Library location is unchanged. (If a library other than the one in which the template resides is specified, this instruction is ignored.)	New template will be placed in the user library and not under parent template, if no Builder rule is violated.
in a library being imported	Library's name must be specified	- Existing template is replaced by imported template. - If imported template's parent differs from that of the existing template, the parent is reset.	- Template is imported and placed in the library specified. - If the library does not exist, it is created.

6. Database Administration

6.10. Import TPS Builder Objects

When the template is...	Import Selection	Results of Import	
		Object is in Target Repository	Object is Not in Target Repository
for strategy(ies) being imported	Project is specified	- Existing template is replaced by imported template. - Library location is unchanged. - If imported template's parent differs from that of the existing template, the parent is reset.	- Template is imported, and placed directly under its parent template. - Imported template's parent is created, if it does not exist in target repository.
For strategy(ies) in a project being imported	Project is specified	- Existing template is replaced by imported template. - Library location is unchanged. - If imported template's parent differs from that of the existing template, the parent is reset.	- Template is imported, and placed directly under its parent template. - Imported template's parent is created, if it does not exist in target repository.

Results of strategy import

A strategy may be imported by itself or as part of a project.

When a strategy is imported, associated files, such as a file inserted into a drawing or an attached program, are also imported. In terms of the strategy itself, results of an import are explained in the following table.

When the strategy is...	Import Selection	Results of Import	
		Object is in Target Repository	Object is Not in Target Repository
itself being imported	Project is specified	- Existing strategy is replaced by imported strategy. - Project location is unchanged.	New strategy is created from parent template and placed in specified project.
in a project being imported	Create new project	Not Applicable	New strategy is created from parent template and placed in specified project.
	Existing project is specified	Existing strategy is replaced by imported strategy.	

Templates are also affected when a strategy is imported. Refer to “Results of template import” for information.

Results of library import

A library can be imported. If Help has been attached to the library, it will also be imported. In terms of the library itself, results of an import are explained in the following table. Also explained is what occurs to a library as the result of other import activities.

When the library...	Import Selection	Results of Import	
		Object is in Target Repository	Object is Not in Target Repository
Is itself being imported.	None	Existing library is replaced by imported library.	New library is created.
Contains a template being imported.	Any library selection.	Existing library is not affected, other than the replacement of an existing template or addition of a new template within it (if this library is the one specified for the import).	Not applicable (an existing library must be selected for the import of a template).
Contains a template from which a strategy being imported was created.	A project is specified.	Existing library is not affected, other than the replacement of an existing template or addition of a new template within it, as required for the imported strategy.	No new library is added; required templates are added to the library containing the template's immediate parent.
Contains a template from which a strategy in a project being imported was created.	A project is specified.	Existing library is not affected, other than the replacement of an existing template or addition of a new template within it, as required for the strategy in the imported project.	No new library is added; required templates are added to the library containing the template's immediate parent.

Templates are obviously affected when a library is imported. Refer to the previous subtopic, "Results of template import."

Results of project import

A project can be imported. Results of an import are explained in the following table.

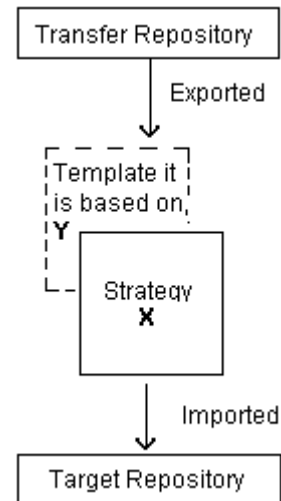
When the project...	Import Selection	Results of Import	
		Object is in Target Repository	Object is Not in Target Repository
Itself is being imported	Create New	Not applicable	A new project is created.
	Project of same name is specified.	Existing project is replaced by imported project.	Not applicable
	Different project is specified.	Not applicable.	Not applicable.
Contains a strategy being imported	A project is specified.	The strategy in the project is replaced by the imported strategy. The project is otherwise unaffected.	Not applicable (a project is specified for the import)

Import example

Consider the adjacent figure. You export strategy X, which was created from template Y.

The results of your import depend on what is already in the target repository, and what your overwrite options are:

- If X and/or Y did not exist in the target repository, then they are added.
 - X is added to the project you specified, and
 - Y is added to the library containing its parent.
- If X existed and you chose to overwrite it,
 - X is imported and overwrites the X in the target repository.
 - If Y always exists, then Y will also be overwritten.
- If X existed but you chose not to overwrite it, then no import of either the strategy or the template would take place.



Plan the import

Planning is the key to a successful import. Considerations include:

For the...	Consider this...
TPS Builder version	• The version of TPS Builder importing objects must be the same as the version that created the transfer repository. To check versions, choose Help > About TPS Builder. • If the version of TPS Builder importing objects precedes the version that created the transfer repository, update TPS Builder before proceeding.
Object(s)	• Do objects by the same name already exist in the target repository? If so, do you want to replace them? (If not, you may not be able to import them, depending on which type of object it is and where you want them to reside.) To check imported objects against your current TPS Builder repository, refer to "Evaluating Export and Import Details" in this document. • Do you want to set up any user libraries or projects in the target repository to receive the objects? • Locate the transfer repository, which is the TPSB_Transfer folder containing the objects to be imported. Make sure the transfer repository is accessible by the system on which the TPS Builder into which objects are being imported
Drive	If you will be importing a large number of objects, ensure the drive containing the receiving TPS Builder repository has sufficient free disk space. The minimum space required for even a small import is approximately 11 MB.
Log	• Decide whether you will overwrite the Export/Import log. This log is accessed by choosing Reports > Logs > Export/Import. If you do not overwrite the log, then all new import activity will be added to the log.
Time	• Consider time requirements. Depending on your system, it may take several minutes to import a small library or project. Ensure you have sufficient time available for the import.



PLANNING TIP

The Export/Import log can become very large. You may find that it is easiest to use the log if you overwrite the log with each export or import.

6. Database Administration

6.10. Import TPS Builder Objects

Import objects procedure

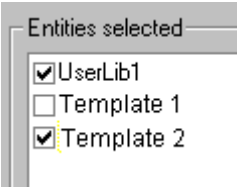
After planning for the import is completed, follow these steps to import TPS Builder objects (libraries, templates, projects, and/or strategies, plus associated files):

Step	Action
Beginning Steps	
1	Start the TPS Builder that is to import objects.
2	Choose File > Import .
3	In the Import (Step 1) dialog box, Folders/Drives list, navigate to the TPSB_Transfer folder.  TIP - Double-click a drive or folder symbol, or name, to make the contents visible in both lists. Clicking the plus sign (+) will not do this. - Do not click in the Contents list. It can only be viewed, not used for navigation.
4	In the Folders/Drives list, click the TPSB_Transfer folder so that the Transfer.mdb file is visible in the Contents list, as shown here:   ATTENTION Optional objects, such as Fieldbus templates, may be in a different folder structure. If so navigate until you see the Transfer.mdb file in the Contents list.
5	Click Next .

Step	Action
Import (Step 2)	
6	In the Import (Step 2) dialog box, evaluate the Objects selected list to ensure the objects listed include those you want to import. If you want to compare this list with the target repository, refer to the topic, "Evaluating Export and Import Details" in this section.
7	In case files attached to or associated with the objects being imported have duplicate names, the suffix "_clone" will be added to file names. If you want a different indicator, • In the Resolve ambiguity section, type the desired word or phrase. • Click Prefix or Suffix as desired. Any characters acceptable for Windows XP file names can be used.
8	If strategies are being imported, select the project into which the strategies should be imported. Note: If you are importing strategies whose names match those of strategies already in your TPS Builder repository, you may choose • a project containing the same strategies, in which case they will be overwritten, or • another project, in which case you will have two different projects containing strategies that are different, but have the same name. If templates are being imported, select the library into which the templates are to be imported. Note: If you are importing templates whose names match those of templates already in your TPS Builder repository, you may choose • a library containing the same templates, in which case they will be overwritten, or • a user library, in which case you will have two different libraries containing templates that are different, but have the same name. • "Default," in which case TPS Builder will check for templates having the same name and will also check the GUIDs, and if they exist, overwrite them with the imported templates. If no templates with the same name exist, TPS Builder will place the imported templates in the library containing the template's immediate parent.
9	Click Next > .

6. Database Administration

6.10. Import TPS Builder Objects

Step	Action
Import (Step 3) - Preview	
10	In the Import (Step 3) - Preview dialog box, review the Objects selected list.
11	If desired, select or clear individual objects in this list. In the adjacent figure, UserLib1 and Template 2 are selected; Template 1 is not. To clear or select an individual object, click in its checkbox. A checkmark indicates the object is selected. To select or clear all objects in the list, click the Select All or UnSelect All button, respectively. 
12	To evaluate details about a specific object, use the Source Repository section of this dialog box, as explained in "Evaluating Export and Import Details" in this section. Also refer to this topic for instructions on comparing objects you are importing to those in the target repository.
13	Select overwrite options: Overwrite Objects: For all objects selected for importing, if objects by the same name exist in the same project or library structure, replace them with the imported object. Overwrite Export/Import Log: Replace the contents of the Export/Import log with the results of the current import activity.  TIP If you are importing a large number of objects, and do not want to take the time to determine whether or not they exist in the current repository, do not select Overwrite Objects. Objects that do not exist in the current repository will be created, but those that do not exist will not be replaced. The Import/Export log will contain details of the import action for your evaluation.
14	When the Import (Step 3 – Preview) dialog box is completed as desired, click Next .
Import (Step 4) - Summary	
15	In the Import (Step 4) dialog box, review the details of the import. If they are correct, click Finish to start the import. A progress indicator shows the status of the import.
16	When the import is completed, an acknowledgement message appears. Click OK .

Cancel an import

If you must cancel an import while it is in progress, follow these steps:

Step	Action
1	In the progress indicator, click Cancel .
2	In response to the confirmation message, click Yes .
3	In response to the cancellation notice, click OK .

Import problems

In some cases, an import or cancellation thereof may fail. In this case, you should delete the temporary folder created by the import, in order to increase your disk space. To do so, follow these steps:

Step	Action
1	Quit TPS Builder.
2	Using Explorer or My Computer, navigate to the HWIAC/databases/TPS Builder/ folder (or equivalent location, if you did not use the default installation).
3	Locate the Temp folder, and in it, the Transferx folder.
4	Delete the Transfer folder (and its contents).

If imported objects appear in your TPS Builder, and you do not want them, delete them using normal TPS Builder procedures. Refer to the *TPS Builder User Guide* for delete instructions as follows:

Section	Topic
Using Libraries	Basic Library Procedures, <i>Delete a user library</i>
Using Templates	<i>Deleting Templates</i>
Working with Projects	<i>Deleting Projects and Project Contents</i>

6.11 Evaluating Export and Import Details

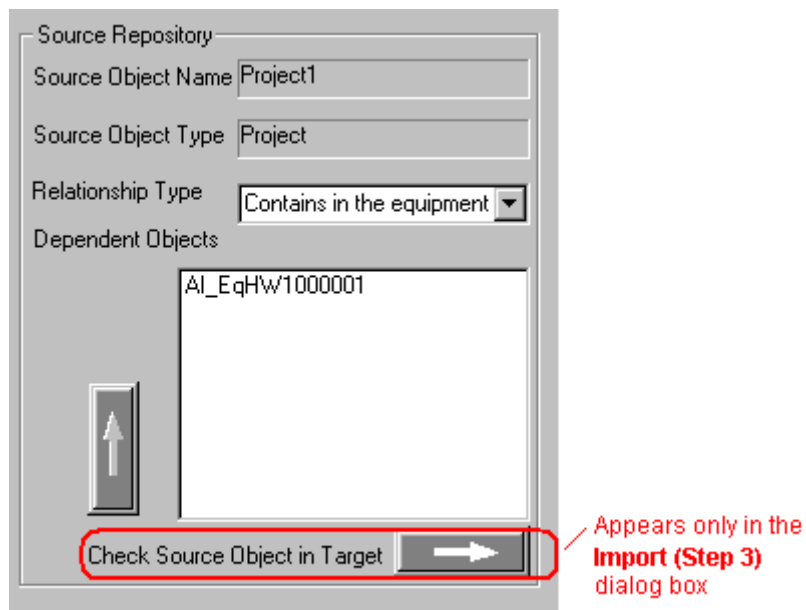
Description

Whether you are exporting or importing TPS Builder objects, you can obtain details about the objects. In **Export (Step 2)** and **Import (Step 3)**, tools in the dialog boxes help you evaluate the characteristics of an object, including relationships with other objects.

Details on using these tools are presented in this section.

Evaluate source objects

Information can be evaluated in the **Source Repository** portion of the **Export (Step 2)** or **Import (Step 3)** dialog box, as shown here:



To evaluate source objects, follow these steps:

Step	Action
1	Do one of the following: - Complete steps 1 - 5 of the export procedure. In this section, refer to Export TPS Builder Objects, "Export objects procedure," or, - Complete steps 1 - 9 of the import procedure. In this section, refer to Import TPS Builder Objects, "Import objects procedure."
2	In the Objects selected list, click the object of interest.
3	In the Source Repository portion of the dialog box, note the Relationship Type being presented. If desired, choose another Relationship Type .
4	Evaluate the resulting Dependent Object information. (Only Information relevant to export/import will be displayed, and not all object-related information)
5	For more information on specific dependent strategies or templates, click the object of interest in the Dependent Objects list. The Up Arrow button becomes active, as shown here.
6	Click the Up Arrow button. Information about the selected dependent object is now presented.
7	Evaluate the information. If desired, make another selection (another relationship type or dependent object), and continue evaluating relationships.
8	When finished, continue with your export or import.



Comparing transfer and target repositories

When you import TPS Builder objects, you can compare objects in the transfer (source) repository and the target (current) repository using the **Check Source Object in Target** (right arrow) button in the **Import (Step 3) - Preview** dialog box, as shown in "Evaluate source objects."

6. Database Administration

6.11. Evaluating Export and Import Details

Comparison criteria

The **Check Source Object in Target** (right arrow) button is enabled whenever the object selected in the **Objects selected** list is available in target repository. For example,

- **Library or project:** If you have selected a library or project and a library/project with same name exists in target repository, then the button is enabled.
- **Templates:** If the selected template is available in any of the libraries of target repository, the button is enabled.
- **Strategies:** If the selected strategy is already available in the target project (that you selected for the import), then the button is enabled.

Comparison procedure

Follow these steps to compare objects in the transfer and target repositories. After reading through the procedure, refer to the following “Comparison example.”

Step	Action
1	Complete steps 1 - 9 of the import procedure. In this section, refer to Import TPS Builder Objects, “Import objects procedure.”
2	In the Objects selected list, click the object of interest. If the selected object exists in the current repository, then the Check Source Object in Target (right arrow) button will be enabled (black instead of gray), as shown here.
3	Click the right arrow to compare the source and target objects. The Dependencies dialog box appears with pertinent information.
4	Evaluate the details shown in the Dependencies dialog box.
5	If desired, evaluate further details about dependent objects in either the transfer or the target repository: 1) Click the object of interest. 2) Click the up arrow.
	The selected object will become the source object in this dialog box, and you can now examine transfer and target contents based on the new source.
6	When you are finished examining the source and transfer relationships, close the Dependencies dialog box by clicking the X in the upper right corner (or pressing the Esc button).
7	If desired, select another object for comparison and repeat the procedure.
8	When finished making comparisons, complete the import, beginning with step 10 of Import TPS Builder Objects, “Import objects procedure” in this section.

Comparison example

Assume you are importing a library, UserLib1, and you want to compare it with the contents of the target repository. After you complete steps 1 - 9 of the import procedure, you will be in the **Import (Step 3) - Preview** dialog box, the pertinent section of which appears here:

Entities selected

- ☒ UserLib1
- ☒ Template 1
- ☒ Template 2
- ☒ Template 3
- ☒ Template 4

Select All UnSelect All

☒ Overwrite Export/Import Log

Source Repository

Source Object Name UserLib1

Source Object Type Library

Relationship Type Library contains Root Tem

Dependent Objects

- Template 1
- Template 2
- Template 3
- Template 4

Check Source Object in Target

To compare the library, click its name in the dialog box, then click the **Check Source Object in Target** (right arrow) button. The **Dependencies** dialog box appears.

The upper portion of this dialog box shows the object you selected, its type, and offers you a choice of applicable relationships to examine:

Source Object Information

Source Object Name UserLib1

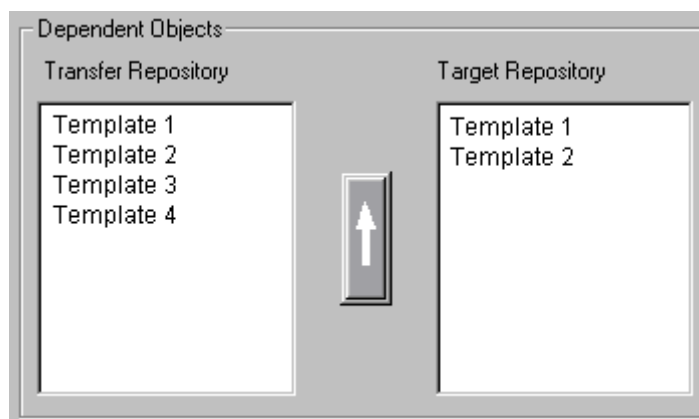
Source Object Type Library

Relationship Type Library contains Root Templates

6. Database Administration

6.11. Evaluating Export and Import Details

The lower portion shows the related objects (based on relationship type) in both transfer and target repositories:



In this example, the transfer repository contains four templates, two of which have the same name as templates in the target repository. If you complete the import of UserLib1 and all of its templates, Templates 1 and 2 will be overwritten and Templates 3 and 4 will be added in the target UserLib1.

6.12 Migrate TPS Builder Repository

Overview

You can migrate an earlier version of a TPS Builder repository ("R130 or Later" or "Pre-R130") to the current version. A new repository will be created that contains all of the data from the old repository. Run-time migration will no longer be possible.

Permissions

You must have administrator privileges on the Repository server to execute this function.

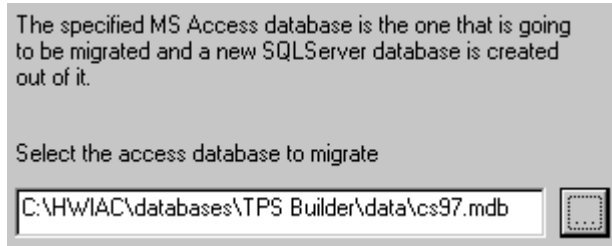
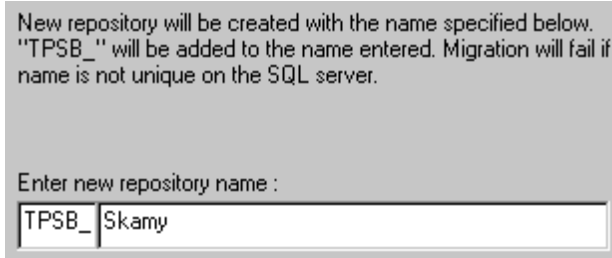
Procedure

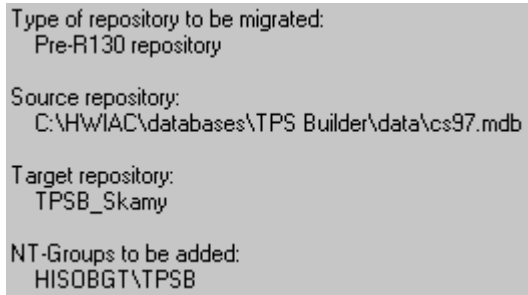
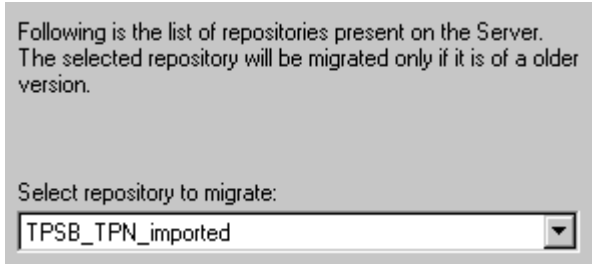
To migrate a TPS Builder repository, execute the following steps:

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed. (If you do not have administrator privileges, an error message explaining that will be displayed and you will not be able to continue.)
5	In the right pane, click the Migrate Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears.
7	Select the radio button to specify the type of repository that is to be migrated, "R130 or Later" or "Pre-R130," and click Next . Another dialog box appears.
8	If you selected the R130 or later repository, go to Step 19. If you selected the Pre-R130 repository, continue with Step 9.

6. Database Administration

6.12. Migrate TPS Builder Repository

Step	Action
	Pre-R130 Repository Path
9	The dialog box that appears after selecting the Pre-R130 repository contains a data input port similar to the following one:  The specified MS Access database is the one that is going to be migrated and a new SQLServer database is created out of it. Select the access database to migrate C:\HWIAC\databases\TPS Builder\data\cs97.mdb
10	Using the Browse button (...), navigate to the .mdb file for the Pre-R130 repository you want to migrate and click Next. Another dialog box appears, containing a data input port for the name of the new repository, with the first 5 characters of the name set to TPSB_. (If the path you specified in the previous step was invalid (doesn't point to an existing, Pre-R130 TPS Builder database), an error message will appear.)
11	Enter the remainder of the new repository name. Here is an example:  New repository will be created with the name specified below. "TPSB_" will be added to the name entered. Migration will fail if name is not unique on the SQL server. Enter new repository name : TPSB_ Skamy
12	Click Next. The current SQL server is checked to see if the new repository name you entered already exists. If it does an error message stating that the selected repository already exists.
13	If the repository name you entered was unique, a dialog box appears, listing the default logins that are to be added to the migrated repository. Enter the Domain Name and NT Group/User for the group/user that you want to be able to access the migrated repository.
14	If you want to remove a login from the login list, select it and click Remove.
15	Click Add. The data entered in the previous step will appear in the List of logins.

Step	Action
16	Repeat Steps 13 through 15 until you have added/removed all the desired login data.
17	Click Finish. A Pre-R130 confirmation dialog box appears, listing the source repository path name, the new (target) repository name, and the NT Groups to be added. Here is an example: 
18	If the data shown is correct, click OK to complete the migration. If the data shown is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
	R130 or Later Repository Path
19	The dialog box that appears after selecting the R130 or later repository in Step 7 contains a combo box, listing all of the TPS Builder repositories on the current SQL server. Here is an example: 
20	Select the R130 or later repository you want to migrate and click Finish .

6. Database Administration

6.12. Migrate TPS Builder Repository

Step	Action
21	If the selected repository not a valid TPS Builder repository, or if it is not a later version, an error message will be displayed. If the selected version is a later version, an R130 or later confirmation dialog box appears, listing the name of the repository to be migrated. Here is an example: Type of repository to be migrated: R130 or later repository Name of repository to be migrated: TPSB_TPNOOnly
22	If the data shown is correct, click OK to complete the migration. If the data shown is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.

6.13 Create TPS Builder Repository

Overview

You can create a new TPS Builder on the TPS Builder repository server. This new repository will be a current version, the same as the one you get when you install the TPS Builder package. All users, whose login was specified during the creation of the new repository, will be able to use this newly created repository.

Permissions

You must have administrator privileges on the SQL server to execute this function.

Procedure

To create a TPS Builder repository, execute the following steps:

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed.
5	In the right pane, click the Create Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears, containing a data input port for the new repository name, with the first 5 characters of the name set to TPSB_.
7	Enter the remainder of the new repository name and click Next . Another dialog box appears.
8	Enter the Domain Name and NT Group for the login to be added.
9	Click Add . The login data just entered will appear in the List of logins to be added.
10	Repeat steps 8 and 9 until you have added all the desired logins.
11	Click Finish . A confirmation dialog box appears, listing the data you have input in the previous steps.
12	If the data is correct, click OK to complete the Create procedure. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
13	A progress bar displaying the database creation completion percentage is displayed. If desired, click Cancel to terminate the Create procedure.

6. Database Administration

6.14. Delete TPS Builder Repository

6.14 Delete TPS Builder Repository

Overview

You can use the Delete Repository function to delete a TPS Builder repository, including the database and the repository directory under the TPS Builder root directory.

Permissions

You must have administrator privileges on the SQL server to execute this function.

CAUTION

Because there is no way that the Delete Repository function can determine if a repository is a database for an offline user, you must check with all other users to make sure that it is OK for you to delete the repository before you actually delete it.

Procedure

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the Administrator Utilities icon. Icons for each Administration Utility function are displayed.
5	In the right pane, click the Delete Repository icon. A dialog box listing some important information about this function is displayed.
6	Click Next . Another dialog box appears, containing a combo box for displaying the existing TPS Builder repositories.
7	Locate and then select the repository you want to delete.
8	Click Finish . A confirmation dialog box appears, listing the SQL server name, and the name of the repository you want to delete.
9	If the data is correct, click OK to complete the delete procedure. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to any previous page and change your input.
10	Progress bar displaying the percentage of the database deleted is displayed. If desired, click Cancel to terminate the delete procedure.

6.15 Switch TPS Builder Repository

Overview

When you start up TPS Builder, it connects to the TPS Builder repository you used last. Using the Switch Repository function, you can switch the current TPS Builder repository.

Procedure

Step	Action
1	On the taskbar, select Start > Programs > TPS Builder .
2	Click the TPSB-AT icon.
3	From the tree view on the left pane, open the TPS Builder Administration Tool folder.
4	In the right pane, click the General Utilities icon. Icons for each General Utilities function are displayed.
5	In the right pane, click the Switch Repository icon. A dialog box listing some important about this function is displayed.
6	Click Next . Another dialog box appears, containing a combo box for displaying the existing TPS Builder repositories.
7	Locate and then select the name of the repository you want to switch to.
8	Click Finish . A confirmation dialog box appears, listing the SQL server name, and the name of the repository you want to switch.
9	If the data is correct, click OK to complete the switch function. If the data is not correct, click Cancel to go back to the last page of the Wizard, where you can go back to each of the previous pages and change your input.
10	If the repository selected is invalid or the database needs to be migrated, an error message will be displayed.

6. Database Administration

6.15. Switch TPS Builder Repository

7. TPS Builder API

7.1 Overview

APIs

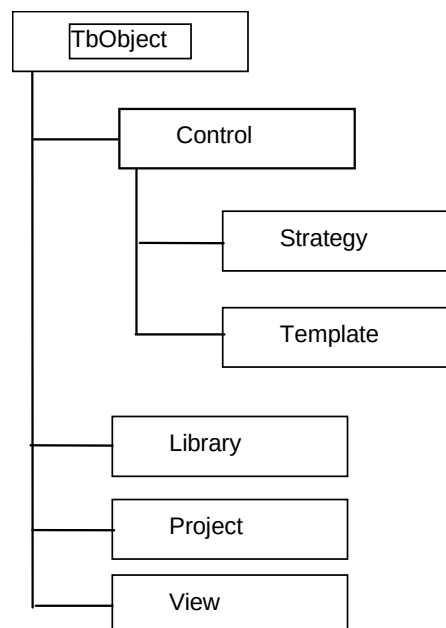
An Application Programming Interface (API) allows software products like TPS Builder to expose their functionality to scripting tools and other applications. TPS Builder exposes an OLE Automation API.

Intended use

The TPS Builder API is intended for users proficient in scripting.

Class hierarchy

This diagram shows a class hierarchy. Methods and properties defined on an object at a higher level in the hierarchy can be used on objects at a lower level. For example, the Name property, which is defined on TbObject, can be used on a Strategy object.



7.2 Examples

Example use

Consider the following scripting scenario:

1. A CreateObject call creates a control library (or TPS Builder API) Lib object.
2. On the Lib object, you can invoke the LogonCtrlSession method. This returns a CCtrlSession object.
3. From the CCtrlSession object, you can call an OpenLibraryByName method, returning a Library object.

You can also call an OpenProjectWithName method, which returns a Project object.

These objects, their methods, their signatures (the parameters they require and the value being returned) and related Help information are available from the Object Browser used in applications like Microsoft Excel. The Help information is in tpsbapi.hlp, a standard Windows Help file.

Example API script

One of the applications that can take advantage of the TPS Builder API is Microsoft Excel (95 or later version). An example script showing how to import data from an Excel spreadsheet into TPS Builder is provided with TPS Builder, in the product under the Examples folder.

To access this example,

Step	Action
1	If you have not created the TPS Builder project into which you want the Excel data to be imported, do so from the base TPS Builder application.
2	Start Microsoft Excel.
3	Open ExcellmpTool.xls and ExcellmpData.xls from the folder in which you installed TPS Builder, such as ProgramFiles\Honeywell\TPS\Tools\Builder\Examples. The files' contents are as follows.

Excel File	Contents
excelimptool	Contains the Visual Basic for Automation (VBA) script for importing an Excel spreadsheet into TPS Builder. This code is driven from the controls on the worksheet. The controls are shown in the following figure.
excelimpdata	Contains some sample data in sheet1 and sheet2. Both can be used with the script.

TPS Builder Import

Project:

Workbook:

Worksheet:

☒ Overwrite existing entities

Note:
This utility will not handle logic points, device control points or node configuration

Step	Action
4	In Excel, select Window > ExcellImpTool.
5	Click the Initialize Lists button. This action retrieves the list of projects in the TPS Builder database and the list of open workbooks in Excel. It fills in the Workbook and Worksheet controls with default values. After initialization, the status bar displays "Initialization done."
6	In the Project field, select the TPS Builder Project into which you want to import data.
7	If the default entries in the Workbook and Worksheet fields are not correct, select the workbook and worksheet from which you want to import data.
8	Click the Import Data button. Note that the status bar displays "Starting import...", "Processing <point name>...", and "Import done."

7. TPS Builder API

7.3. Using the Script

Note that if you leave TPS Builder open while doing this import, with the project window open, the TPS Builder project window will not be updated. You must refresh the window by pressing Ctrl+F5 to see the result of the import.

7.3 Using the Script

Using the script with your data

If you have TPS Process Network (Process Manager family or Application Module) implementation data in an Excel spreadsheet, you can use the ExcelImpTool.xls script to import the data into a TPS Builder project. All entities except logic points, device control points, and node entities are supported.

To use the script with your own data, follow these guidelines:

- Parameter data need a header = TPS Network parameter name.
- Parameter data need either a PNTTYPE field or a LIBRARY and TEMPLATE field for importing new strategy blocks. Existing strategy blocks need only a name. If both are provided, the point type will be used to look up the template to be used for import.
- Using the Library/Template field, you can select different templates for points with the same point type. If you provide only point type, the code will do a lookup in the table in the spreadsheet. This means you can still specify which template you want to use for a given point type, but only one per point type.
- Data can be organized in rows or columns.
- You can have zero, one, or more spaces between the parameter name and the parameter data.
- Data can start anywhere in the in the area A1:Z26, that is, header data, like general project information, will be skipped.

Viewing the script

To view the script,

Step	Action
1	Start Excel.
2	Open ExcellmpTool.xls from the folder in which you installed TPS Builder, for example, Program Files\Honeywell\TPS Builder\Examples. For a description of this file, see step 3 in "Example API script".
3	From the View menu, choose Toolbars > Visual Basic to show the Visual Basic toolbar.
4	Click the Visual Basic Editor button,  .
5	If necessary, click the Project Explorer button,  to view the tree.
6	From the Project Explorer tree, the code is located in: • Microsoft Excel Objects > SheetImport, and • Modules > ErrorMsg
7	Click the Object Browser button,  .
8	In the resulting dialog, select Lib from the upper combo box that shows <<All libraries>>.
9	Now the lower combo box will list all the TPS Builder objects.
10	When selecting one of these objects, the list on the right will show all the methods and attributes on the object.
11	When selecting one of these methods or attributes, the signature of the method will show up at the bottom (list of parameters required for using the method, the types of the parameters and the return value).
12	Clicking the Help button,  , displays the TPS Builder API Help for the method or attribute. This online Help provides descriptions, syntax, return values, and examples for the API classes.

7. TPS Builder API

7.3. Using the Script

Creating strategies from external data

Different approaches for creating strategies from external data exist using a) TPS Builder backbuild (discussed beginning in the *Importing TPS Entity Data from TPS Files (Backbuilding), Overview*) section of the *TPS Builder Configuration Tool User's Guide* or b) the import tool discussed in this section. The approaches depend on what is being created:

Step	Action
1	To use one template per point type (no template modules): a) Perform a standard backbuild. b) Perform a standard import.
2	To use several templates per point type (no template modules): a) Backbuild in groups, switching to the appropriate template for each group. b) Import using the Library/Template field (refer to "Using the script with your data.")
3	To use template modules: a) Create strategy modules in TPS Builder, entering as a minimum, only the names for each strategy block. b) Backbuild or import data to fill in the parameters for the strategy blocks.

8. Maintaining Adequate Disk Space

8.1 Overview

Microsoft Windows recommends the following actions to help you solve problems with disk space. Note that defragmenting requires that you have a defragmentation tool installed.

- Defragment a disk or volume. Do this after the following steps, to maximize the value of defragmenting.
- Empty the Recycle Bin.
- Reduce space used by the Recycle Bin.
- Back up unneeded files to tape or copy them to a floppy disk and remove them from your hard disk.
- Search for and remove temporary files (don't forget those "Cache" directories created by frequent Internet use).
- Search for and remove files larger than a certain size.
- Remove Windows components that you don't use.
- Create more disk space by using disk compression on NTFS volumes. Do not compress the TPS Builder database folder, however. (Performance will degrade.)

Refer to Windows Help for details on these procedures. From the taskbar, choose **Start > Help**.

8. Maintaining Adequate Disk Space

8.1. Overview

Honeywell

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA